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In-Depth Analysis of Coal Demand Dynamics in Tajikistan until 2050



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Acronyms

µg – Microgram

ADB – Asian Development Bank

CAPEX – Capital Expenditure

CBAM – Carbon Border Adjustment Mechanism

CCS – Carbon Capture and Storage

CHP – Combined Heat and Power

COP – Conference of the Parties

Gcal – Gigacalorie

GDP – Gross Domestic Product

GHG – Greenhouse Gas

GW – Gigawatt

GWh – Gigawatt-hour

HELE – High Efficiency, Low Emissions

HPP – Hydro Power Plant

IEA – International Energy Agency

INDC – Intended Nationally Determined Contribution

ktCO₂eq – Thousand Tonnes of Carbon Dioxide Equivalent

kW – Kilowatt

kWh – Kilowatt-hour

LCOE – Levelized Cost of Electricity

LLC – Limited Liability Company

LULUCF – Land Use, Land-Use Change and Forestry

MtCO₂Eq – Million Tonnes of Carbon Dioxide Equivalent

MW – Megawatt

NDC – Nationally Determined Contribution

OJSC – Open Joint-Stock Company

OPEX – Operational Expenditure

RES – Renewable Energy Sources

SPP – Solar Power Plant

SUE – State Unitary Enterprise

TJ – Terajoule

TJS – Tajikistani Somoni

TPP – Thermal Power Plant

TWh – Terawatt-hour

UE – Unitary Enterprise

UN – United Nations

UNECE – United Nations Economic Commission for Europe

UNFCCC – United Nations Framework Convention on Climate Change

USAID – United States Agency for International Development

USD – United States Dollar

VAT – Value-Added Tax

WPP – Wind Power Plant

Executive summary

At first glance, one would expect the demand for coal in the energy sector in Tajikistan to increase, as the country has experienced power shortages (recognized by the state-owned energy holding company “Barki Tojik” in 2022) in the face of rising energy demand.

The current government plans to increase generation capacity by upgrading existing hydro power plants and constructing new solar and wind power plants are insufficient to cover the shortfall in electricity demand (these plans do not include completion of the Rogun HPP due to uncertainty over the sources of financing for this project).

One possible solution to the electricity shortage would be to build a new coal-fired power plant. The decision seems rather natural, given the country’s rich coal reserves. Furthermore, the construction of the coal-fired power plant fits perfectly with the “*National Development Strategy of the Republic of Tajikistan for the period until 2030*”, adopted in 2016 and with the “*Concept of the Development of the Coal Industry in the Republic of Tajikistan until 2040*”, adopted in 2019. Both documents aim to increase coal production to 10.4 million tonnes in 2030, 5 times higher than the annual average production level in 2019-2022. However, such growth in coal supply will not be possible unless it is backed by an adequate growth in demand.

This report presents the results of a comparative analysis of the levelised cost of electricity (LCOE) of new coal, gas, solar, wind and hydro power plants.

As calculations have shown, in the case of a new power plant financed by Tajikistan, a coal-fired power plant appears to be the most competitive option under the minimum LCOE criterion. The result is due to 3 factors:

- the low cost of local coal;
- the low capital cost of building a new power plant, thanks to access to relatively cheap Chinese technology; and
- the high (10 percent) discount rate used in the developing country contexts for budget financing of investment projects.

Even taking into account the costs associated with the damage to public health that air pollution resulting from operations of a coal-fired power plant would cause, this option still remains the most effective.

The above assessment, however, does not take into account the cost of greenhouse gas emissions due to the absence of explicit carbon pricing in Tajikistan.

The construction of the coal-fired power plant (even with a capacity that would completely cover the electricity deficit once all the planned projects are implemented) would also not prevent Tajikistan from meeting commitments stated in the country’s nationally determined contribution (NDC).

All above notwithstanding, despite the apparent attractiveness of the coal option, **it is very unlikely that any new coal-fired power plants will be built in Tajikistan**, due to the lack of available funding sources.

The state-owned energy holding company Barki Tojik will not be able to finance the construction of a coal-fired power plant due to its high indebtedness. Private companies are also unlikely to be interested in building such power plant due to low electricity tariffs that do not allow to recover the cost of generation. The option of obtaining loans for such an endeavour from international banks does not appear probable either, as in 2021 many countries and organizations announced that they would stop financing “unabated” coal projects. For Tajikistan, a similar decision made by China is of particular importance, as that country seemed to be the most likely to invest in any potential coal-related energy projects in Tajikistan. The previous coal-fired power plant, commissioned in 2014, was 95% financed by a subsidized loan from the Chinese state-owned Exim bank.

In addition, when selecting projects for financial support, institutions use a significantly lower discount rate (the so-called “social” rate). Such rate reflects the value of the investment to society, which is interpreted much more broadly than just in terms of financial return.

In this paper, the social discount rate has been taken as 3%. At this rate, the cheapest option for closing the electricity deficit is to complete constructing the Rogun HPP rather than to build any new power plant.

Therefore, coal demand from the energy sector in Tajikistan is not expected to increase (nor is, at the same time, an existing coal-fired power plant expected to close before its scheduled lifetime).

In industry, the only obvious prospects for an increased demand for coal are in cement production, in which coal will remain the most cost-effective fuel for a considerable period of time.

The aggregate demand of the remaining consumers, i.e. households and public institutions, will remain virtually unchanged by 2050 (as compared to 2021 levels).

The projects using coal as feedstock (e.g. in the iron and steel industry, or in coal-to-olefins conversion processes) should not be totally discounted though. The availability of cheap coal in the country might be an important component of the potential success of such projects. However, as it is only one of many factors that need to be taken into account, any project of that type should be subject to the most thorough and objective feasibility study. Currently, there are no adequately justified projects of that kind.

The situation in world markets in 2022 was extremely favourable for Tajikistan’s coal exports, which reached the record 372 000 tonnes level. However, it should be understood that circumstances governing at that time were exceptional, and therefore such conditions can be by no means treated as the “new normal”. As long as the price situation remains favourable, coal exports will continue. At the same time any decisions on investments in export infrastructure (e.g. railway construction) or in increasing coal production capacity should not be made solely on the basis of current economic trends in the sector, without evaluating the real prospects of the global coal market, which is characterized by a decline of both: demand and price.

As a result, the goals for increasing coal production indicated in the current strategic documents of Tajikistan seem to be outdated and appear to be unlikely to be achieved.

The focus of a new coal strategy, if any such document is to be developed, should not be on achieving record production levels, but rather on ensuring the equitable distribution of natural rents from coal exports, safe working conditions, minimizing environmental damage from operating, reclaiming and closing coal mines, and proper management of methane emissions from coal mines after the mining ceases.

Introduction

The current *“Concept of the Development of the Coal Industry in the Republic of Tajikistan until 2040”* assumes that Tajikistan’s coal production in 2030 will reach 10.4 million tonnes. This is five times more than the average annual production between 2019 and 2022.

The target production level of 10.4 million tonnes in 2030 was adopted in 2016 and since then it has never been revised, even though in 2020 actual coal production was half of what was envisaged for that year.

The relevance of the *Concept* has come into question due to the fact that since its adoption there have been significant changes in both the domestic and foreign coal markets.

An important domestic circumstance was the resumption in 2018 of natural gas supplies to Tajikistan from Uzbekistan, the cessation of which six years earlier had dramatically increased Tajikistan’s interest in coal as an energy source, stimulating the construction of the country’s first coal-fired power plant in Dushanbe.

The past 5-6 years have seen a major shift in global climate policy towards an accelerated phase-out of fossil fuels. Tajikistan supports major international agreements in this area, and unlike some neighbouring countries, it has not yet set its climate neutrality targets. At the end of 2022 the country adopted a *“Strategy for the Development of “Green” Economy in the Republic of Tajikistan for 2023-2037”*, which in its concept differs markedly from the country’s coal strategy.

This report responds to the question of the extent to which the current strategy of coal sector development is consistent with today’s realities. The discussion is based on the authors’ estimates of coal demand from the energy sector, industry, and the population of Tajikistan until 2050.

Sections 1-5 describe the actual energy balance of Tajikistan, present the current situation of the domestic coal market in the country, and analyse its current coal strategy.

Sections 6-9 are dedicated to the forecast of domestic coal demand in different segments of Tajikistan’s economy and of coal exports from the country until 2050, as well as an analysis of prospects for substituting coal as a domestic source of energy production with alternative solutions. Finally, the document offers conclusions and recommendations which follow from the analysis.

1. Energy needs of Tajikistan: status and baseline forecast

1.1. Energy consumption in Tajikistan in 2015-2020

The structure of energy consumption in Tajikistan is characterized by a high share of hydropower (see **Table 1.1**). HPPs account for more than 90% of electricity generation (see **Figure 1.1**).

Table 1.1 - Primary energy consumption in Tajikistan by source, 2015-2020, TJ

Source	2015	2016	2017	2018	2019	2020
Hydro	61 200	59 875	61 680	66 220	69 012	65 210
Crude oil and petroleum products	41 814	42 245	36 746	44 356	46 427	46 997
Coal	25 928	31 312	32 250	36 752	39 384	35 775
Natural gas	156	122	45.1	2 208	9 024	9 024
Firewood	-	43 436	-	-	-	-
Other solid biofuels	-	11 156	-	-	-	-

Source: *Tajstat*

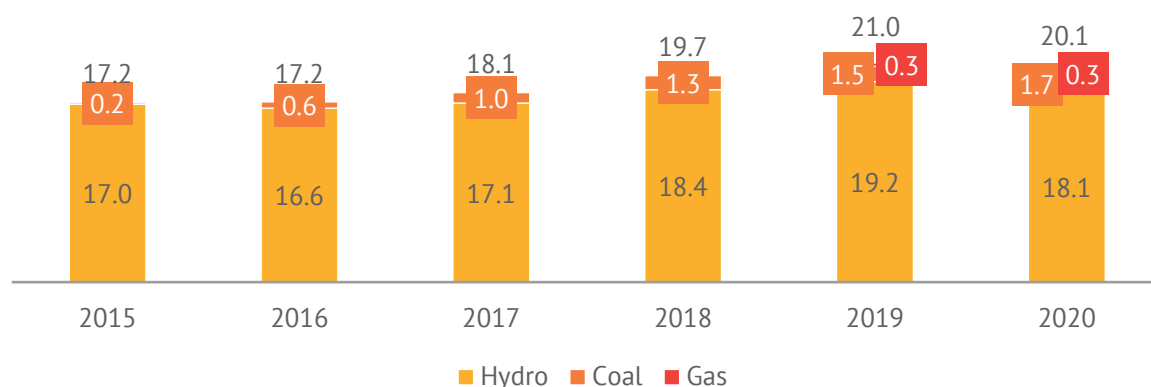
The share of firewood and other solid biofuels (manure, maize and cotton stalks) in energy consumption is also high, but the statistics on that matter are limited to 2016, and the data are derived by extrapolating the results of a one-time survey of households, individual entrepreneurs and businesses on “*The state of the energy facilities and efficiency of use in 2016*”¹.

Given that firewood and other solid fuels are mainly used for heating, there is a reason to believe that demand for them has decreased in recent years, as Dushanbe CHP-1 reopened at the end of 2018. This CHP plant, using natural gas from Uzbekistan as a fuel (and low-sulphur fuel oil as a backup fuel), produces both heat for the district heating system in the Tajik capital Dushanbe as well as electricity that is used by households for heating.

In addition, the first temporary unit of the Rogun HPP was launched in 2018 and the second one in 2019 (each with capacity of 120 MW), which increased generation of electricity that can also be used for domestic heating.

¹ Tajikistan Statistical Agency (2018), Report on the results of one-off sample survey on “The state of the energy facilities and efficiency of use in 2016”, URL: <https://www.stat.tj/en/news/publications/report-on-the-results-of-one-off-sample-survey-on-the-state-of-the-energy-facilities-and-efficiency-of-use-in-2017->

Figure 1.1 - **Gross electricity production by source, 2015-2020, TWh**



Source: data provided by a national consultant

1.2. Problems of energy security in Tajikistan

Ensuring energy security and efficient use of electricity is one of the strategic goals outlined in the “*National Development Strategy of the Republic of Tajikistan for the period until 2030*”, adopted in 2016.² Achieving that goal entails, as stated in the above strategy, “establishing an effective system of risk management and monitoring of energy security, including unrestricted and equal access to energy resources for all consumers”.

One of the key problems hindering the achievement of “unlimited and equal access to energy resources for all consumers” in Tajikistan is the recurring shortage of electricity in the autumn-winter period. Once shortages occur, electricity supply to certain households (usually in rural areas of the country) is restricted, preventing them not only from using electric appliances, but also electric heaters and thus forcing the use of coal, wood, and other solid biofuels for heating.

As indicated by the results of the above-mentioned household survey, “*On the state of the energy economy...*” such a situation was observed, in particular in 2016. At that time, one of the reasons for the electricity shortage was the termination of natural gas supply from Uzbekistan, which lasted from 2012 to 2018, and put the Dushanbe CHP-1 out of operation.

The halting of gas supplies from Uzbekistan prompted the accelerated construction of the Dushanbe TPP-2 operating on local coal, the first phase (two units with a combined capacity of 100 MW) of which was commissioned on 10 November 2014, and the second (two more units with a combined capacity of 300 MW) on 8 December 2016.

In the absence of natural gas supplies, industrial enterprises in Tajikistan also began to switch to coal, primarily in the cement industry.

However, despite the partial substitution of coal for gas in power generation and industry, as well as restoration of natural gas supplies from Uzbekistan and the reopening of Dushanbe CHP-1 in 2018, the risk of electricity shortages during the cold season has still not been eliminated. Recently, on 13 October 2022, the state-owned power holding “Barki Tojik”³ announced a forced limitation of electricity supply for the autumn-winter period and estimated the seasonal electricity deficit at 1 billion kWh.

² Text of the document in English, URL: https://nafaka.tj/images/zakoni/new/strategiya_2030_en.pdf

³ Text of the document in Tajik, URL: <http://barqtojik.tj/tj/info/news/197716/>

The main reasons for electricity shortage in the autumn-winter period are: (a) a sharp seasonal increase in electricity demand due to mass use of electric heaters by households, and (b) limited electricity production at hydropower plants due to seasonality of water flow. In the above-mentioned statement of “Barki Tojik”, it is noted that the flow rate in the Vakhsh River, where most of the operating Tajik HPPs are located (including the largest one – Nurek HPP with 3GW capacity and Rogun HPP with 3.6 GW capacity, currently under construction), is 10 times lower in winter (150–200 cubic meters per second) than it is in summer (2 000–2 200 cubic meters per second).

The seasonal drop in water flow in the Vakhsh is partially compensated by its advanced accumulation in the Nurek reservoir. In 2021, for example, the cumulative gross production of electricity in the 6 cold months (January–March and October–December) was only 23% lower than in the 6 warm months (April–September) (see **Table 1.2**).

Table 1.2 - Gross electricity production by hydropower plants in Tajikistan in 2021, TWh

Indicator	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Gross electricity production	1.6	1.2	1.3	1.2	1.8	1.9	2.1	1.9	1.8	1.1	1.3	1.5
% of annual production	9%	7%	7%	6%	9%	10%	11%	10%	9%	6%	7%	8%

Sources: *Barki Tojik*, *Tajstat*

Another implemented solution to prevent winter electricity shortages is a seasonal reduction of exports: in 2021, the autumn-winter period (January-February and October-December) accounted for less than 10% of total annual net electricity exports from Tajikistan (see **Table 1.3**).

Table 1.3 - Export, Import, and Net Export of Electricity from Tajikistan in 2021, GWh

Indicator	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Exports	154	111	98	54	439	482	596	567	523	76	95	108
Imports	107	71	72	31	63	92	88	99	44	59	72	86
Net exports	47	40	26	23	375	390	508	468	478	18	24	22

Sources: *Tajstat*

The operation of Dushanbe CHP-1 and CHP-2 is also seasonal (only in autumn and winter). During that time, they provide additional electricity and heat production, which reduces the demand for electricity that is used for heating purposes by Dushanbe residents.

The above measures allow for an increase of electricity supply to the domestic market during the autumn-winter period. For example, in 2021 electricity supply to the domestic market was 9.6 TWh from January to March and from October to December compared to 8.6 TWh from April to September (see **Table 1.4**).

Nevertheless, these measures have been shown to be insufficient to ensure unrestricted access to electricity during the autumn-winter period.

Table 1.4 - Production (by source), net export and domestic electricity consumption in 2021, GWh

Source	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Hydro	1 621	1 218	1 315	1 189	1 767	1 854	2 059	1 948	1 761	1 120	1 328	1 549
Gas (Dushanbe CHP-1)	69	55	62	31	0	0	0	0	0	12	40	60
Coal (Dushanbe CHP-2)	235	201	216	117	0	0	0	0	0	106	219	233
Net export	-47	-40	-26	-23	-375	-390	-508	-468	-478	-18	-24	-22
Domestic electricity consumption	1 878	1 434	1 567	1 314	1 392	1 464	1 552	1 480	1 282	1 220	1 563	1 821

Sources: *Barki Tojik, Tajstat*

The solution to this situation seems to be the expansion of national generation capacity. Given that almost all natural gas consumed in the country (over 99%) is imported from Uzbekistan (and being mindful of the precedent of Uzbekistan unilaterally cutting off those supplies), the solution most consistent with the Tajikistan's national security objectives would be to use its domestic resources for electricity generation, i.e. coal, hydropower and renewable energy.

1.3. Baseline forecast of energy demand in Tajikistan until 2050

Neither the above-mentioned “*National Development Strategy of the Republic of Tajikistan until 2030*” nor other policy documents contain a forecast of energy demand. Given these circumstances, the international consultant hired by UNECE⁴ performed the demand forecast. The forecast was based on the methodology used to develop a baseline scenario for greenhouse gas emissions of Tajikistan under the Fourth National Communication⁵ of Tajikistan to the UN Framework Convention on Climate Change.

That methodology assumes that energy demand is linked to proxy variables. In this study it was assumed that households' energy demand increases in proportion to the population growth, while other consumers' demand increase proportionally to GDP growth, adjusted for the reduction in the energy intensity that was projected by extrapolating the historical development of this indicator for the years 2000-2020. Forecasts were electricity demand and final demand for other energy sources, such as coal, natural gas, and petroleum products (i.e. demand for them excluding for the purpose of transforming into electricity). Demand for firewood and other solid biofuels was not projected due to lack of sufficient statistics.

The UN Department of Economic and Social Affairs Population Division's projection of 2022 was used as a population projection.

The GDP forecast of Tajikistan was also carried out by the international consultant. The forecast for the period 2022-2025 was based on the forecast of the Ministry of Economic Development and Trade of the Republic of Tajikistan, for the period 2025-2030 on the forecast from the industrial scenario of

⁴ Mr. Ivan Khomutov, with support of the company that he is the General Director of, namely Petromarket RG

⁵ Text of the document in Russian and English, URL: <https://unfccc.int/documents/614376>

the “*National Strategy*”. After 2030, a gradual decline in real GDP growth rates was assumed, based on the historical dynamics of real GDP in Tajikistan and the global trend towards a decline in GDP growth rates of developing countries.

The GDP, population, and energy intensity reduction rate projections adopted in this research are presented in **Table 1.5**.

Table 1.5 - Forecast of real GDP, population, and rate of reduction of energy intensity of GDP in Tajikistan until 2050

Indicator	Fact	Forecast					
	2020	2025	2030	2035	2040	2045	2050
Real GDP at 2021 prices, bln TJS	91.9	133.4	191.6	262.9	350.2	459.1	595.7
Average annual growth rate (CAGR) of real GDP, %	3.7% (2015-2020)	7.8% (2020-2025)	7.5% (2025-2030)	6.5% (2030-2035)	5.9% (2035-2040)	5.6% (2040-2045)	5.3% (2045-2050)
Population at the end of the year, million	9.72	10.52	11.42	12.36	13.35	14.32	15.2
Average annual growth rate (CAGR) of population, %	2.1% (2015-2020)	2.0% (2020-2025)	1.7% (2025-2030)	1.6% (2030-2035)	1.6% (2035-2040)	1.4% (2040-2045)	1.2% (2045-2050)
Average annual rate of reduction (CAGR) of energy intensity of GDP, %	-1.9% (2015-2020)	-2.3% (2020-2025)	-2.2% (2025-2030)	-2.0% (2030-2035)	-2.0% (2035-2040)	-1.9% (2040-2045)	-1.9% (2045-2050)

Sources: UN Department of Economic and Social Affairs Population Division, Petromarket RG, based on data provided by a national consultant

The forecast for energy demand in Tajikistan until 2050 is presented in **Table 1.6**.

Table 1.6 - Forecast of energy demand in Tajikistan until 2050

Product	Unit of measure	Fact		Forecast					
		2015	2020	2025	2030	2035	2040	2045	2050
Electricity	GWh	13 901	16 420	19 553	23 453	27 590	32 006	36 807	42 022
	TJ	50 044	59 110	70 391	84 430	99 325	115 221	132 503	151 279
Including household demand, taking into account the 1000 GWh deficit	GWh	5 914	7 857	8 504	9 235	9 998	10 798	11 577	12 292
	TJ	21 290	28 285	30 614	33 247	35 993	38 874	41 677	44 249
Coal	thousand tonnes	789	1009	1264	1587	1930	2298	2705	3156
	TJ	14 976	18905	23687	29741	36182	43075	50695	59154
Including household demand	thousand tonnes	283	181	196	213	231	249	267	284
	TJ	5 367	3 400	3 680	3 996	4 326	4 673	5 010	5 319

Product	Unit of measure	Fact		Forecast					
		2015	2020	2025	2030	2035	2040	2045	2050
Natural gas	mln m ³	4	105	136	175	217	261	311	366
	TJ	156	4 212	5 435	6 993	8 653	10 431	12 410	14 624
Crude oil and petroleum products	thousand tonnes	975	1 064	1 151	1 250	1 354	1 462	1 567	1 664
	TJ	41 814	46 997	50 867	55 242	59 804	64 592	69 249	73 523

Source: Petromarket RG based on the data provided by Tajtstat

It is worth noting that this projection is essentially a baseline scenario, or in other words a projection into the future of the current consumption pattern. It takes into account only the dynamics of the country's social and economic development indicators and does not reflect potential structural changes in fuel consumption – e.g. a possible decrease in coal demand from the population due to an increase in electricity available for heating; an increase in coal demand due to the construction of a new coal-fired power plant; a decrease in demand for petroleum products due to an increase in electricity demand resulting from an increase in number of electric vehicles, etc. The forecast presented in **Table 1.6** serves as a reference point to project coal demand until 2050 under different industry development scenarios.

An alternative coal demand forecast has been provided by the national consultant. It is based on a different approach to demand forecasting: it considers potential coal consumers (including external market) that could meet demand in line with the country's coal production growth plans. This projection is presented in Section 2.

2. The energy strategy of Tajikistan

2.1. Key strategic objectives for energy

At the moment, there is no documented energy strategy with a horizon until 2050 in Tajikistan. Official strategic/programmatic documents touching on energy issues have a shorter time horizon. Such documents are: “*National Development Strategy of the Republic of Tajikistan until 2030*” (adopted in 2016), “*Concept of the Development of the Coal Industry in the Republic of Tajikistan until 2040*” (adopted in 2019), and “*Accelerated Industrialisation Programme of the Republic of Tajikistan 2020-2025*” (adopted in 2020).

The “*National Development Strategy*” sets out the main strategic objectives in the energy sector: ensuring the country’s energy security and the efficient use of electricity. The achievement of these goals implies the following tasks:

- Diversification of energy sources in power generation, envisaging the development of hydropower resources of large and small rivers, development of existing oil and gas and coal capacities, development of new fossil fuel deposits, creating of technical opportunities for the use of non-conventional (renewable) energy sources (solar, wind, biological, geothermal), modernization of existing and construction of new HPPs and TPPs;
- Efficient use of available energy capacity and realization of the export potential of the electricity sector;
- Modernization and technical upgrading of the oil and gas industry, development of new oil and gas fields;
- Widespread energy conservation and improved energy efficient of the national economy;
- Development of internal and external energy infrastructure (power grids and substations);
- Establishment of an effective energy security risk management and monitoring system, including unrestricted and equal access to energy resources for all consumers;
- Ensuring financially viable and sustainable operation of the energy sector; and
- Integrated water resources management.

These tasks should result in the development of the country’s electricity sector on the basis of the “*10/10/10/10 concept*”, namely:

- Increasing the design capacity of the electricity system to 10 GW;
- Increasing annual electricity exports to neighbouring countries to 10 billion kWh;
- Ensure diversification of the country’s electricity system capacity by at least 10%, by increasing generation capacity from other energy sources, including coal, oil, gas and renewable energy sources; and
- Reduction of electricity losses in the country by up to 10%.

In addition, in accordance with the “*National Development Strategy*” up to 500 million kWh of electricity is to be saved by introducing energy-saving technologies and improving energy efficiency.

2.2. Strategic objectives for the development of the coal industry

Regarding the coal industry, the “*National Strategy*” sets target (indicative) coal production volumes for 2020, 2025 and 2030 in two scenarios – (1) industrial, and (2) industrial-innovative. The “*Coal Industry Development Concept*” reproduces the coal production target for 2030 from the industrial scenario of the “*National Strategy*” and supplements it with the target for 2040 (all these targets are

indicated in **Table 2.1**). Comparison of the target coal production volumes for 2020 from the “*National Strategy*” with actual production in 2020 (see **Table 2.1**) shows that the coal industry development rates set by the “*National Strategy*” currently are not met.

The more recent “*Programme for Accelerated Industrialisation of the Republic of Tajikistan 2020-2025*” gives a more realistic short-term forecast, which shows coal production in 2025 well below the target levels set in the “*National Strategy*” (see **Table 2.1**).

Table 2.1 - Targets for coal production in Tajikistan until 2040 (compilation of data from various strategic documents) and actual coal production in 2015 and 2020, million tonnes

Source	2015	2020	2025	2030	2040
National Development Strategy of the Republic of Tajikistan, Industrial Scenario, 2016	1.0 (fact)	4.1 (target)	6.9 (target)	10.4 (target)	-
National Development Strategy of the Republic of Tajikistan, Industrial-Innovative Scenario, 2016	1.0 (fact)	5.3 (target)	10.3 (target)	15.1 (target)	-
Concept for the development of the coal industry, 2019	-	-	-	10.4 (target)	15.0 (target)
Accelerated Industrialisation Programme of the Republic of Tajikistan 2020-2025, 2020	-	2.1 (target)	2.4 (target)	-	-
National statistics	1.0 (fact)	2.0 (fact)	-	-	-

Sources: National Development Strategy of the Republic of Tajikistan until 2030, Tajikistan Coal Sector Development Concept until 2040, Accelerated Industrialisation Programme of the Republic of Tajikistan 2020-2025, data provided by the national consultant

The “*Concept of Coal Industry Development*” explains this trend: “Analysis of the last two years shows that due to increased imports of natural gas from a neighbouring country and the commissioning of the Rogun hydropower plant, the purchase of coal by the population and a number of industrial enterprises has decreased”. On that basis, the “*Concept of Coal Industry Development*” concludes that “special attention should be paid to the problems of production of other coal products and increase of coal exports”.

In addition to the coal production targets contained in the strategy/programme documents, three coal demand projection options have been presented by Ministry of Industry and New Technologies of the Republic of Tajikistan and SUE Angishti Tojik (see **Table 2.2**)

Table 2.2 - Forecast of coal demand (including export)/production, in Tajikistan (2 options), thousand tonnes

Option	Fact			Forecast					
	2020	2021	2022	2025	2030	2035	2040	2045	2050
Option 1	2 096	1 943	2 769	3 820	8 700	10 300	13 000	13 700	15 000
Option 2	2 096	1 943	2 769	3 800	8 100	9 500	12 000	12 800	14 000

Source: Ministry of Industry and New Technologies of the Republic of Tajikistan, SUE Angishti Tojik

In these projections, coal demand includes exports and is balanced with expected production. In this case, the figures in **Table 2.2** can be interpreted as both demand and production projections. The details of the scenario conditions under which the presented coal demand/production projections are valid, are not known.

Additionally, Ministry of Industry and New Technologies of the Republic of Tajikistan and SUE Angishti Tojik also provided the demand forecast by a consumer category for Option 1 (see **Table 2.3**).

The general forecast of the potential demand for coal from different consumer categories is presented in the second part of the report.

Table 2.3 - Forecast of demand for coal mined in Tajikistan by consumer category, Option 1, thousand tonnes

Consumers		Period					
Title	Capacity	2025	2030	2035	2040	2045	2050
Energy:							
Dushanbe CHP	400 MW	1 100	1 100	1 100	1 100	1 100	1 100
Huaxing Guyur Cement CHP	25 MW	60	60	60	60	60	60
Fon Yagnob CHP	600 – 700 – 1 050 MW	-	1 500	2 000	2 200	2 200	2 200
Shurab CHP	300 MW	-	550	650	650	650	650
Industry:							
Cement production	7 mln tonnes (by 2025)	630	693	762	839	922	1 015
Coke production	1 000 tonnes	-	1 500	1 500	3 000	3 000	3 000
Anthracite production	300 tonnes	-	600	650	650	650	650
Other industrial enterprises	-	450	500	550	600	600	600
Public sector organisations (heating)	-	80	100	120	120	120	120
Population (heating)	-	500	400	480	600	430	300
Export	-	1 000	1 700	2 400	3 100	3 800	5 000
Total		3 820	8 703	10 272	12 919	13 532	14 695

Source: Ministry of Industry and New Technologies of the Republic of Tajikistan, SUE Angishti Tojik

3. External conditions for the development of the coal market in Tajikistan

3.1. International environmental pressure and the obligations of Tajikistan in that field

Tajikistan ratified the UN Framework Convention on Climate Change (UNFCCC) in 1998 and the Kyoto Protocol in 2008, thus accepting its obligation as a Party to the Convention.

In October 2015, Tajikistan submitted its estimated Intended Nationally Determined Contribution (INDC)⁶ for the implementation of the relevant decisions adopted by the 19th and 20th Conferences of the Parties (CoP) to the UNFCCC.

In 2017, the Parliament of the Republic of Tajikistan ratified the Paris Agreement to the United Nations Framework Convention on Climate Change.

In October 2019, a National Climate Change Adaptation Strategy for the period until 2030 was adopted by Decree of the Government of the Republic of Tajikistan.⁷

In October 2021, the country prepared an updated version of the nationally determined contribution (NDC)⁸. Under the new NDC, the country plans to adhere to an unconditional target and, if financial and technical support are received, to a conditional target as well.

The updated NDC assumes an unconditional target of limiting GHG emissions (including LULUCF) to 60-70% of 1990 levels by 2030 (21.32–24.87 MtCO₂Eq), and a conditional target of 50-60% of 1990 levels by 2030 (17.76–21.32 MtCO₂Eq).

In 2016 – the most recent year for which GHG emission data are available for Tajikistan – the country's GHG emissions amounted to 13.75 MtCO₂Eq⁹.

In the baseline scenario with no additional GHG emission reductions, assuming that the proportions of fuel consumption for energy (the energy mix) remain at their current level, emissions in 2030 are projected to be 23.54 MtCO₂Eq, which is within the range of the unconditional target.

This shows the relatively low ambition of the Tajikistan's NDC's unconditional targets. At the same time, it also means that increased coal consumption without additional emission reduction measures might result in a failure to meet the unconditional targets set in NDC.

Furthermore, increasing coal consumption without additional measures to reduce emissions from coal use would run counter to the call to reduce consumption of "*unabated coal*" contained in the decisions of the 26th and 27th Conferences of the Parties to the UNFCCC.

Tajikistan has not declared its carbon neutrality targets.

⁶ Text of the document in English, URL: <https://cop23.unfccc.int/sites/default/files/NDC/2022-06/INDC-TJK%20final%20ENG.pdf>

⁷ Text of the document in Russian, URL:

<https://mel.cgiar.org/reporting/download/hash/a8caf00db5684e499a539e248300b2bd>

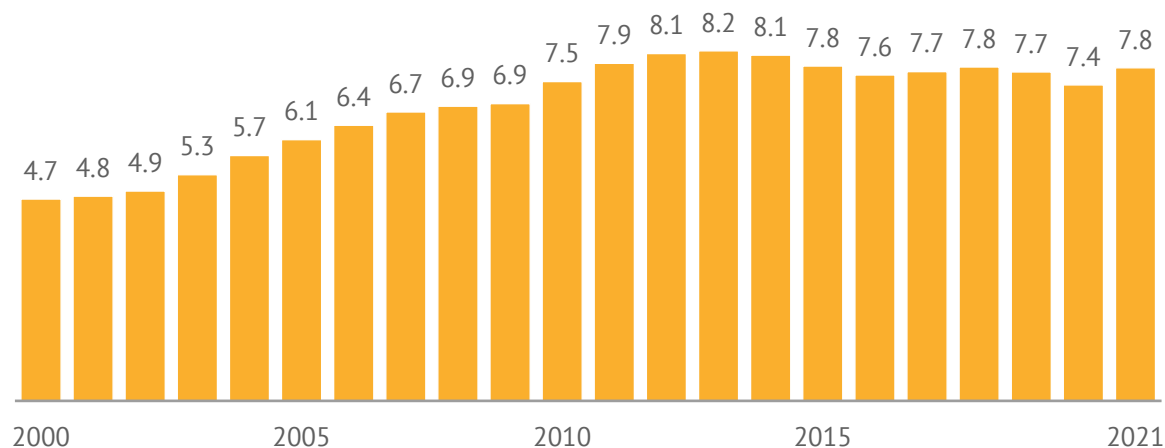
⁸ Text of the document in English, URL: https://unfccc.int/sites/default/files/NDC/2022-06/NDC_TAJIKISTAN_ENG.pdf

⁹ Total GHG emissions are calculated by Petromarket RG on the basis of the GHG emissions and removals inventory tables for 1990-2016 from the Annex to the Fourth National Communication of the Republic of Tajikistan to the UN Framework Convention on Climate Change, using the global warming potential (GWP) values provided by the IPCC in the Fourth Assessment Report (25 to CH₄ and 298 for N₂O)

3.2. Global and Central Asia coal markets

Global coal demand stopped growing in 2013 and has plateaued in recent years at around 7.7 billion tonnes per year (see **Figure 3.1**).

Figure 3.1 - **World coal demand in 2000-2021, billion tonnes**

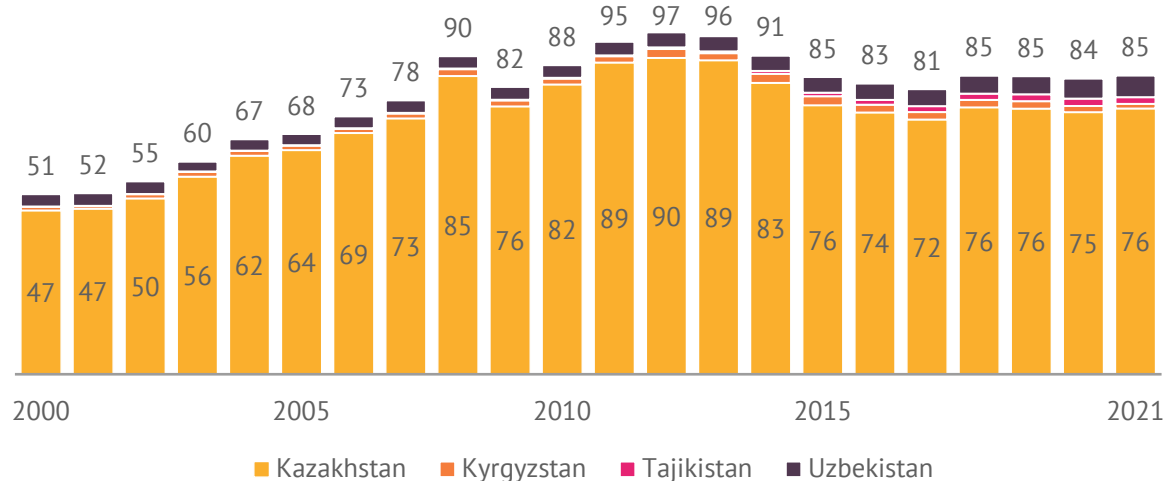


Source: U.S. Energy Information Administration

The International Energy Agency (IEA) estimates that in 2022, as the global economy recovers from the *COVID-19* pandemic and natural gas prices rise, coal demand may surpass the previous record level set in 2013. At the same time, according to IEA, within a few years, demand will start to decline again, and by 2030, even in the Stated Policies scenario that does not imply the introduction of additional emission reduction measures compared to those already in place, it will decrease by 9% as compared to 2021 levels.

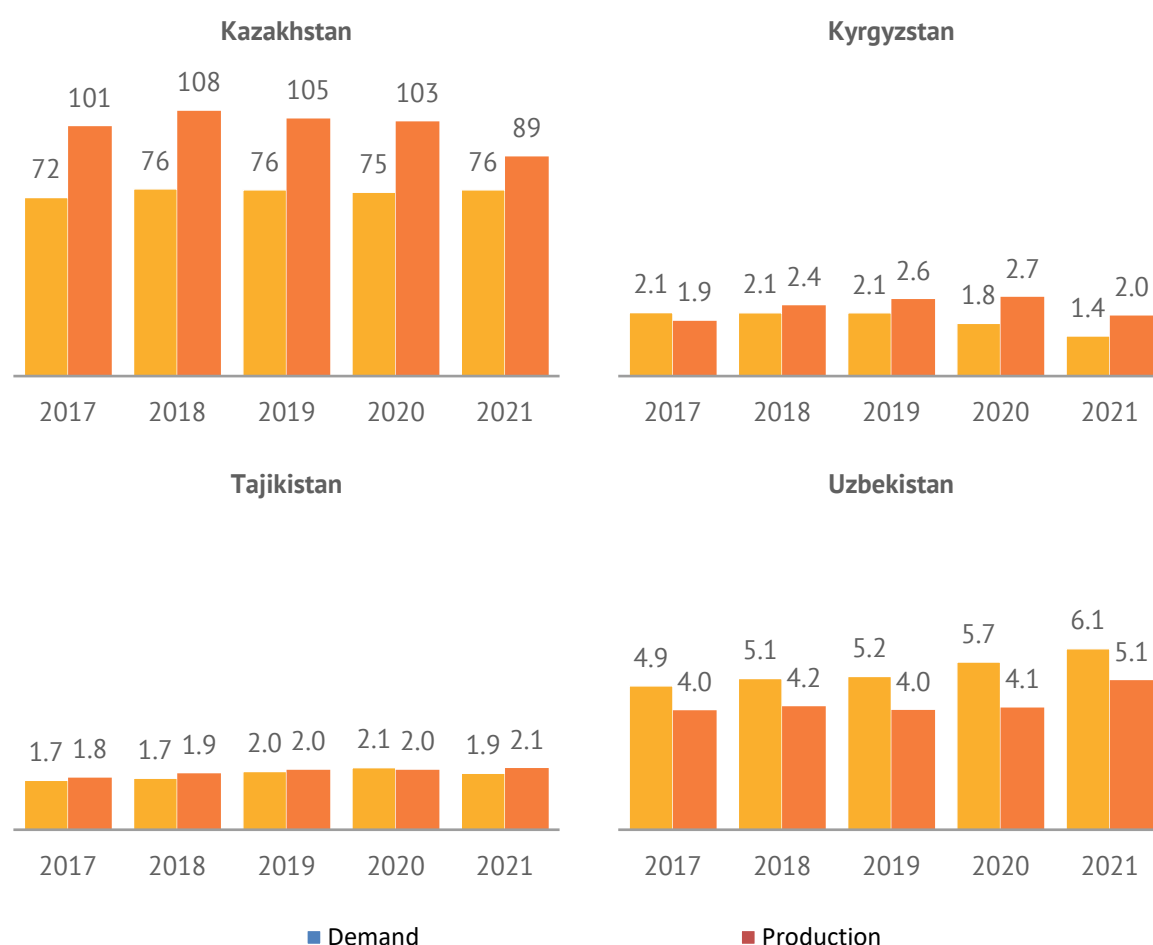
Coal demand in Central Asia in 2021 was 85 million tonnes (1% of world demand), with Kazakhstan accounting for 76 million tonnes, Uzbekistan 6.1 million tonnes, Tajikistan 1.9 million tonnes and Kyrgyzstan 1.4 million tonnes (see **Figure 3.2**). There is no demand for coal in Turkmenistan.

Figure 3.2 - **Coal demand in Central Asian countries in 2000-2021, million tonnes**



Sources: U.S. Energy Information Administration, data provided by a national consultant

Figure 3.3 - Coal demand and production in Kazakhstan, Kyrgyzstan, Tajikistan and Uzbekistan in 2017-2021, million tonnes



Sources: U.S. Energy Information Administration, data provided by a national consultant

3.3. Impact of the current geopolitical situation on coal prices and energy supplies from abroad

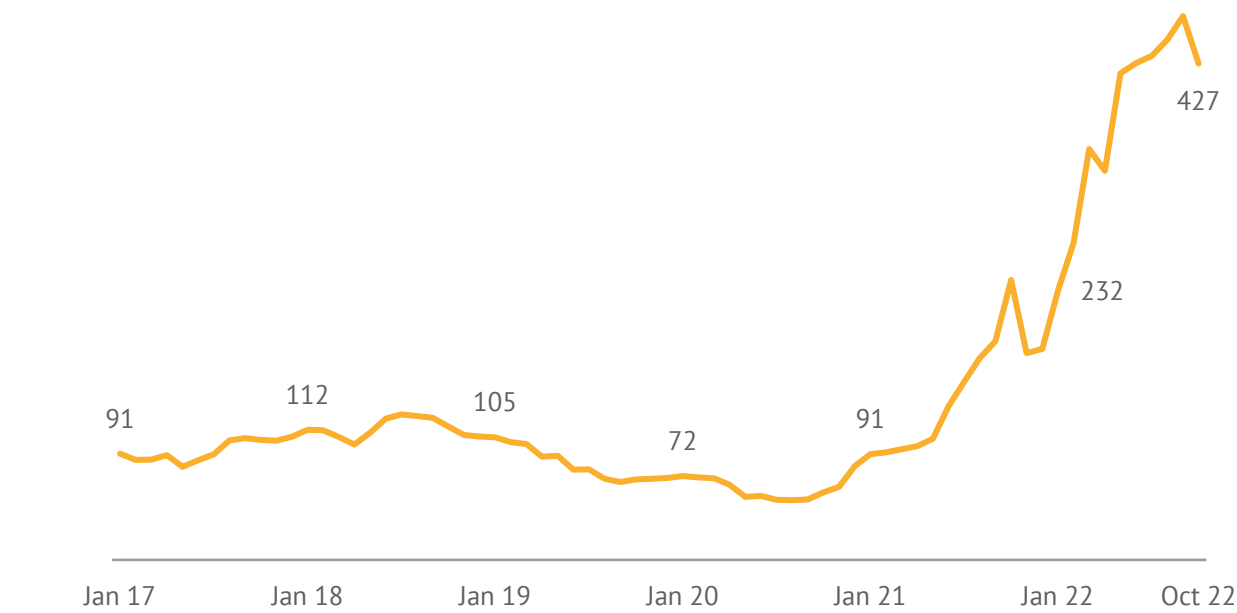
The geopolitical situation of 2022 has led to a significant increase in coal prices on the international market (see **Figure 3.4**), but has had little impact on coal prices in Tajikistan due to the isolation of the local market, in which transport restrictions hinder exports and there is no need for imports.

While the average FOB Newcastle price for coal in the first 9 months of 2022 has increased threefold as compared to the same average price in 2021, over the same period the price of coal in Tajikistan has increased on average by only 7%.

As for the supply of other energy products (natural gas, crude oil and petroleum products), the current geopolitical situation does not pose additional risks: the supply of natural gas from Uzbekistan and petroleum products from the Russian Federation in 2022 continues as normal.

The tense situation on the Kyrgyz-Tajik border in 2022 also does not threaten the energy security of Tajikistan, due to the small imports of energy products from Kyrgyzstan.

Figure 3.4 - Australian steam coal price at FOB Newcastle, 2017-2022, USD/tonne



Source: IMF Primary Commodity Price System

4. Coal market in Tajikistan

4.1. Coal market capacity in Tajikistan

The capacity of the domestic coal market in Tajikistan in 2021 was 1 909 thousand tonnes, roughly the same as in 2019 and 2020 (see **Figure 4.1**).

Table 4.1 - Coal production, demand, exports, imports and stock changes in Tajikistan in 2020-2021, thousand tonnes

Rate	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Production	203	237	412	518	875	1 042	1 362	1 760	1 907	2 028	2 029	2 089	2 400
Export	2	22	2	0	0	0	0	5	43	18	5	34	372
Import	9	15	24	10	12	14	10	6	7	28	53	96	104
Consumption	213	253	448	493	838	997	1 386	1 669	1 741	1 965	2 096	1 909	2 397
Stock change	-3	-23	-14	35	49	59	-14	92	129	73	-18	242	-265

Source: Ministry of Industry and New Technologies of the Republic of Tajikistan

A period of rapid market growth occurred in 2012-2018, during a period when there was no natural gas supply from Uzbekistan. As a result, in 2014 and 2016 respectively, 2 phases of the coal-fired Dushanbe CHP-2 were commissioned. At the same time some industrial enterprises in Tajikistan switched from gas to coal use. In the following years, it was the cement industry that contributed significantly to the growth of the coal market.

Together with the resumption of gas imports from Uzbekistan, the growth of the coal market slowed down as some of the enterprises that had previously switched to coal returned to using natural gas. However, CHP-2 continues to operate and even increased electricity generation and coal purchases, despite the reopening of the gas-fired Dushanbe CHP-1 in 2018.

4.2. Coal supply: production and imports

As noted in the Subsection 3.2, Tajikistan's coal market is generally self-sufficient, with only 96 000 tonnes of coal imported in 2021, which constitute approx. 5% of the country's total coal demand. In previous years, the share of imports was even lower than that (not more than 3%).

Almost all imported coal is lignite from Kyrgyzstan. Lignite is more suitable for cooking on an open fire than bituminous coal, and Tajikistan's lignite mining companies OJSC Angisht and LLC Siyovush F do not fully cover domestic demand for this fuel.

Demand from key coal consumers (energy and industry sectors) is fully met by local production.

At the same time, the design capacity of the existing coal mining enterprises exceeds 3 million tonnes per year, which is one and a half times the current Tajikistan's demand for coal. This opens up opportunities for a potential increase of production in the future (see Section 5).

4.3. Demand for coal in different economic sectors

Coal demand in the power sector (see **Table 4.2**) is driven by the needs of Dushanbe CHP-2, the first phase of which (two units with a combined capacity of 100 MW) was commissioned on 10 November 2014, and the second one (two more units with a combined capacity of 300 MW) on 8 December 2016. Also in 2016, along with the commissioning of the Huaxing Gayur Sugd Plant Ltd, a 25 MW CHP plant was commissioned to ensure the stable operation of the plant. This CHP plant operates all year round and consumes 60 000 tonnes of coal per year.

Table 4.2 - Demand for coal in Tajikistan by consumer category in 2010-2021, thousand tonnes

Category	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Energy sector ¹	-	-	-	38	210	208	446	833	891	1 071	1 206	900	1 022
Industrial sector	60	97	239	277	385	450	500	567	649	591	599	580	752
Public sector organisations	16	15	13	18	42	43	51	56	55	56	57	60	74
Households	128	126	172	150	189	283	378	208	139	220	181	273	445
Import ²	9	15	24	10	12	14	10	6	7	28	53	96	104
Total	213	253	448	493	838	997	1 386	1 669	1 741	1 965	2 096	1 909	2 397

Source: Ministry of Industry and New Technologies of the Republic of Tajikistan

Notes:

1. Since 2016 includes coal consumption at Huaxing Gayur Cement CHP (25 MW capacity) in the amount of 60 000 tonnes/year. 90% of the electricity generated is used for the Huaxing Gayur Cement plant

2. In the data provided, imported coal volumes are not broken down by consumer category. Imported coal is mainly bought by households

The demand for coal from Tajikistan's industry started to grow strongly after gas supplies from Uzbekistan were interrupted in 2012. Soon after 2012, more than 200 industrial enterprises in Tajikistan switched from gas to coal.

Another driver of coal demand has been the development of the cement industry in Tajikistan. From 2010 to 2021, cement production in the country increased 15-fold, and since 2016, Tajikistan has become a net exporter of cement (see **Table 4.3**). Cement plants currently account for more than 90% of coal consumption by the industrial enterprises.

Table 4.3 - Cement production, demand, export and import in Tajikistan in 2010-2021, thousand tonnes

Rate	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Production	287	298	235	384	1 146	1 450	2 001	3 117	3 844	4 202	4 243	4 280	4 300
Export	0	0	0	0	0	1	282	1 023	1 437	1 548	1 312	1 172	1 377
Import	572	687	944	1 081	514	133	100	44	39	21	9	13	25
Consumption	860	986	1 179	1 465	1 660	1 583	1 819	2 138	2 446	2 675	2 940	3 121	2 948

Source: Ministry of Industry and New Technologies of the Republic of Tajikistan

The production capacity of cement plants in Tajikistan reached 5.6 million tonnes/year in 2020. According to the “*Programme for Accelerated Industrialisation of the Republic of Tajikistan in 2020-2025*” cement production in the country should be brought to the level of 7.0 million tonnes by 2025. As a result, along with the growth of the cement industry, the demand for coal in the industrial sector in general will also increase (according to the estimates of Ministry of Industry, in 2025 it will reach 1 million t/year), despite the fact that with the resumption of the natural gas supplies from Uzbekistan that took place in 2018, the demand for coal in other industries has decreased.

Demand for coal in the non-productive commercial sector is not represented in the official statistics of Tajikistan.

The demand for coal from public institutions in 2021 amounted to 60 000 tonnes (3% of total demand). Coal purchased by public institutions is used for heating.

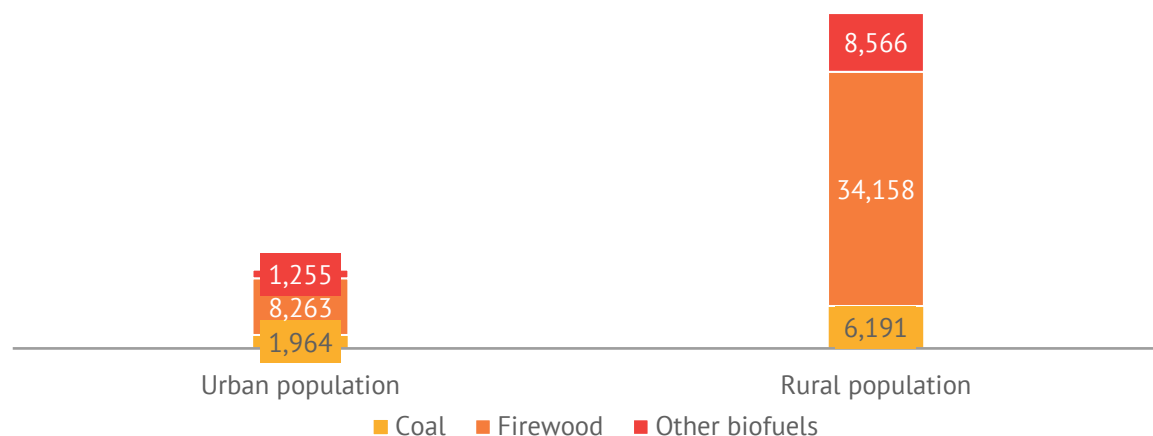
Household demand for coal varies markedly from year to year (see **Table 4.2**): from 139 thousand tonnes in 2018 to 378 thousand tonnes in 2016. This variation is due to differences in weather conditions in different years. Since households use coal for heating and purchase coal in advance of the cold season, in case of a relatively warm winter the population is left with some reserves, which naturally leads to a lower coal demand in the next year.

Data from the “*2016 State of Energy Sector and Energy Efficiency Survey*” gives an alternative estimate of household demand for coal in 2016. It indicates 409.8 thousand tonnes, against 378 thousand tonnes reported by the Ministry of Industry and New Technologies of the Republic of Tajikistan, which are presented in **Table 4.2**. Such discrepancy results from the difference in methodologies used by the two forecasts to derive their respective estimates. While the Ministry generated its data from the producers reports on coal sales, which break down sales by various consumers’ categories, the survey data reflects the consumer side, i.e. it is based on answers to the questions about energy consumption received from 3000 households based in different regions, which were then extrapolated to represent the demand of the entire population of those territories.

The ratio of urban to rural population in coal consumption is about 1:3.

According to the results of the “*The state of the energy facilities and efficiency of use in 2016*,” the main solid fuel used by the Tajik population is firewood rather than coal. The demand of the population for other solid biofuels (manure, maize and cotton stalks, shrubs and wood pellets) is comparable to, but still slightly higher than, the demand for coal (see **Figure 4.1**).

Figure 4.1 - Consumption of coal, wood and other solid biofuels by the population, TJ, 2016



Source: “*The state of the energy facilities and efficiency of use in 2016*”, Tajstat

4.4. Coal efficiency

An important feature of Tajikistan's local coal market is that it is partially regulated.

The domestic market for coal in the country is in fact isolated from the world market. Export of coal to the international market from Tajikistan is hindered by the lack of access to the sea and too high transportation costs via land routes. There is also no need to import coal. As a consequence, local coal prices are not linked to the world prices but are determined by local production costs.

Coal mining companies calculate prices for their products based on the planned production costs increased by a certain rate of return (which varies by enterprise but stays within the 20% range) and then agree to those prices with the state regulator, SUE Angishti Tojik.

Such practice sometimes results in variation of prices in the range of 10 to 20% for the coal of the same origin and quality offered by different mining companies (see Annexes **Annex A**).

Within the country, coal is transported by roads, so a portion of the final logistics costs borne by the consumer can be quite high. For example, Dushanbe CHP-2 purchases coal (mainly from the Fon-Yaghnob mine) at a price of 420 TJS/tonne (37 USD/tonne), of which 120 TJS/tonne (11 USD/tonne), i.e. almost one third, are for logistics.

To make export deliveries enterprises in Tajikistan must have a valid export permit. Therefore, to alleviate the periodically occurring hardship caused by shortage of coal to meet the domestic demand, when necessary, the Government issues a legal ban on coal exports. For example, such ban took place between 2012 and 2016 when gas supply from Uzbekistan was interrupted. Similarly, coal exports were temporarily suspended in 2022 (from 17 August to 12 November) due to the fact that the Dushanbe CHP-2 plant did not stockpile enough coal for the start of the heating season.

5. Assessment of coal reserves and coal production potential in Tajikistan

The coal industry in Tajikistan is characterised by very diverse (1) geographical distribution, (2) mining and geological conditions, and (3) technical equipment of the underground and open pit mines. Mines and open pits are predominantly located in mountainous and foothill areas (with altitude ranging from 1 160 to 4 000 m above the sea level) of the country.

There are 21 registered coal deposits and occurrences in Tajikistan with total resources of 3.6 billion tonnes, including (according to the classification adopted in Tajikistan) 73.4% of inferred resources (categories P1, P2 and P3, 2.656 million tonnes), preliminary estimated reserves 17.8% (category C2, 641.5 million tonnes), and commercial reserves of 8.8% (categories A, B, C1, 317.2 million tonnes). Lignite, bituminous coal (both thermal and coking), and anthracites are mined in the country.

The Tajikistan's coal fields are located in seismically active areas. The main coal deposits and operating coal enterprises are concentrated in the north-western part of the country. In 2021, 80.6% of the production in Tajikistan came from Sughd Region and 19.4% from the Districts under Central Government Jurisdiction. The locations of the largest deposits are shown on the map in **Annex B. Location and characteristics of coal deposits in Tajikistan.**

The deposits are developed by enterprises with a total design capacity of 3 million tonnes/year, and with the actual production of 2 million tonnes in 2021 (**Table 5.1**). Most of the coal is extracted from open pits.

Table 5.1 - Projected capacity and actual production in 2021 of coal enterprises in Tajikistan, in thousand tonnes

Enterprise	Projected capacity, thousand tonnes/year	Development method	Production in 2021, thousand tonnes
OJSC Angisht	300	Underground	91
UE Fon-Yaghnob Mine	800	Open pit	736
	80	Underground	
Talco Resources Ltd.	410	Open pit	456
	86	Underground	
C.M.T. Ltd.	500	Open pit	343
	306	Underground	
UE KA Ziddi	100	Open pit	281
	215	Underground	
Dion Angishti Tojikiston Ltd.	50	Open pit	2
Siyovush F Ltd.	6	Underground	5
UE KA Shuroobod	4	Open pit	0
SHT Sayyod Ltd.	40	Open pit	3
Wuromoon Ltd.	2	Open pit	0

Enterprise	Projected capacity, thousand tonnes/year	Development method	Production in 2021, thousand tonnes
Gunch & Co Ltd.	2	Underground	4
UE KA Nazar-Aylok	15	Open pit	0
Anavac Ltd.	50	Underground	3
Angishti Kamarob Ltd.	90	Open pit	61
Kudrat-2010 Ltd.	-	Open pit	60
Vena Ltd.	-	Open pit	34
Daryo-2015 Ltd.	-	Open pit	10
Total	3006		2089

Source: data provided by a national consultant

The 6 largest companies hold licenses for 98% of the country's coal reserves and account for 94% of total production of Tajikistan (see **Table 5.2**). A complete list of licensees together with the reserves on their balances is provided in **Annex C**. For the annual coal production levels by regions and enterprises see **Annex D**.

Table 5.2 - Share in industrial reserves (as of 1 January 2019) and production of coal enterprises in Tajikistan (in 2021)

Enterprise	Share of reserves on the balance, 1 January 2019	Share in production, 2021
UE Shahgtai Fon-Yaghnob	8.9%	35.2%
Talco Resources Ltd.	28.6%	21.8%
C.M.T. Ltd.	24.1%	16.4%
UE Koni Angishti Ziddi	6.7%	13.4%
OJSC Angisht	26.3%	4.3%
Angishti Kamarob Ltd.	0.2%	3.0%
Others	5.1%	5.9%
Total	100.0%	100.0%

Source: data provided by a national consultant

Coal deposits are developed by both private and state-owned enterprises. In 2021 the share of state-owned companies (OJSC Angisht, UE Shahgtai Fon-Yaghnob, UE KA Ziddi, Angishti Kamarob Ltd.) in the total production was 56%.

The “*Concept for the Development of the Coal Industry of the Republic of Tajikistan until 2040*” identifies 4 priority areas for investment in the development of mining, which aim to increase mining capacity by more than 3 million tonnes/year (see **Table 5.3**).

Table 5.3 - Priority areas for financing the coal sector

Project	Capacity, thousand tonnes/year	Costs, million USD\$	Project to meet the demand for new coal
Construction of mines at the Kante and Jijikrud sections of the Fon Yaghnob deposit	1 200 and 900	160-250	Construction of the 700 – 1 050 MW Fon-Yaghnob TPP will ensure coal demand in the range of 1 676 – 2 514 thousand tonnes/year + construction of the beneficiation and coke-making plant will ensure demand of 1 000 thousand tonnes/year
Reaching design capacity for mine construction at the Rabot site of the Fon-Yaghnob coal deposit	1 000	100	
Construction of a mine in the Garby area of the Ziddi coal field	315 – 1 000	150 – 210	Needs of Dushanbe Combined Heat and Power Plant OJSC
Comprehensive processing of the Nazar-Aylok anthracite deposit, including construction of a processing plant, conditioning and mine construction	No data	100 – 160	Enrichment and subsequent export of anthracite

Source: Concept for the Development of the Coal Industry in the Republic of Tajikistan until 2040

It is important to note that the development of coal mining assumes that additional coal producers will find demand in the domestic market generated by parallel implementation of new projects in the industry and the energy sector. Due to the logistical constraints and the neighbouring countries' low level of interest in Tajik coal, it is usually not possible for the mining companies to export any “excess” volumes of production (2022 being an exception). The availability of the domestic demand is therefore an important factor to be assessed when evaluating the feasibility of investments in the new coal extraction capacity in the country. As a result, most plans aiming to improve coal production are tied to the projects that are designed to increase coal demand and thus have chances to be implemented only if demand increases. At the moment, some reliance on exports is indicated only in the “integrated processing of the Nazar-Aylok anthracite deposit” project (see **Table 5.3**).

The Republic of Tajikistan is planning to carry out geological exploration work in the coal sector until 2025 at a total cost of 6.5 million somoni (\$580,000 USD, see **Table 5.4**).

Table 5.4 - Cost of planned exploration works in the coal sector of Tajikistan, thousand TJS

Exploration work	2021	2022	2023	2024	2025
Dashtichum Coal Exploration	350	700	1 500	1 500	2 000
Margeb Coal Exploration	300	650	1 000	1 000	-
Exploration and feasibility study at the Kaftarkhona site of the Nazar-Aylok anthracite deposit	-	350	1 000	1 500	2 000
Exploration and feasibility study at Ravnov and Sangitang	-	250	1 000	1 000	1 000
Exploration work on the Kafandar, Saimiri, Alisurkhan and Tovasin coal occurrences	-	-	500	1 000	1 500
Total	650	1 950	5 000	6 000	6 500

Source: Ministry of Industry and New Technologies of the Republic of Tajikistan

According to the 2016 emissions inventory, only 0.5% (1.19 thousand tonnes) of methane emissions in Tajikistan are due to methane releases from coal mining. The latter, however, are not measured, but estimated by using emission factors. One of the reasons for the low reported methane emissions is the predominance of open pit coal mining (87% of coal is mined by open pit, while only 13% by underground mining) where, in general, measurements are much harder to perform and much less frequently done (if ever). In terms of the design capacities, 70% of them are associated with surface mining and only 30% with underground extraction.

The coal industry is regulated by the Ministry of Industry and New Technologies. The “*Law on Coal*”¹⁰, which regulates mining activities, was adopted in July 2012.

¹⁰ Text of the document in Russian, URL: <https://medt.tj/documents/main/normativno-pravovie-akti/zakonodatelnie-akti/ru/02544-ru.pdf>

6. Prospects of coal-based power generation in Tajikistan

6.1. Evaluation of additional power demand in Tajikistan

As demonstrated in Sub-Sections 1.2 and 2.1, Tajikistan needs additional power generation, both in the short term (to eliminate a power shortage of 1 billion kWh in autumn and winter) and long term (in 2050, the power demand is expected to grow by 25.6 billion kWh as compared to 2020).

Naturally, this raises the issue of power sources for new generation: is the demand likely to be covered by the fossil fuel-based (coal, gas) or “clean” energy infrastructure (solar, wind, or hydro)?

Before benchmarking response options, it is imperative to keep in mind the seasonal nature of the expected additional power demand: in autumn and winter it is drastically different than in spring and summer. This is important as different power plants can demonstrate different seasonal generation profiles. Solar power plants produce 70% of their total annual generation in spring and summer (April to September) whilst coal-fired power plants are technically capable, should the need be, of being operated only in autumn and winter (as is currently the case of Dushanbe CHP-2).

In order to evaluate the additional power needs of Tajikistan and the available ways to meet them, the brownfield and greenfield power plant projects, both ongoing and proposed, shall be taken into account (for a brief overview, see **Table 6.1**). The list does not include the Rogun HPP completion project, as it requires substantial additional capital investments (approx. 4.2 billion USD) that are not secured yet. Therefore, the Rogun HPP completion project will be discussed separately as an alternative to coal-fired power plants.

The additional power demand forecast was based on an assumption that power losses would decrease within the forecast period: to 10% in 2030 (according to the target of the *National Strategy*, see Sub-Section 2.1), and to 8% in 2050.

Table 6.1 - List of ongoing and proposed projects taking into account in the additional power requirement evaluation for Tajikistan

Project	Expected outcome	Year of completion
Nurek HPP Upgrade	Replacement of all 9 hydropower units, design plant capacity expansion from 3 000 to 3 300 MW	First phase (3 units) — 2023, second phase — 2029
Sarband HPP Upgrade	Replacement of all 5 hydropower units, design plant capacity expansion from 240 to 270 MW	2023
Kairakkum HPP Upgrade	Replacement of all 6 hydropower units, design plant capacity expansion from 126 to 174 MW	2023
Construction of solar power plants	Commissioning of projects with a total capacity of 330 MW	2026-2030
Construction of wind power plants	Commissioning of projects with a total capacity of 130 MW	2026-2030

Source: Petromarket RG, according to the data provided by the Ministry of Power and Water Resources of the Republic of Tajikistan

Table 6.2 shows additional power demand in Tajikistan until 2050 (with a shortage in the counterfactual scenario assuming that no other additional power plants, except the existing and already proposed ones, as mentioned in **Table 6.1**, will be operated).

Table 6.2 - **Power shortage/surplus forecast for the domestic market of Tajikistan until 2050, billion kWh**

Period	2020	2025	2030	2035	2040	2045	2050
Autumn and winter (October to March)							
Net generation	9.1	10.4	11.7	11.7	11.7	11.7	11.7
Transmission losses	1.2	1.5	1.2	1.1	1.1	1.0	0.9
Demand	8.6	10.3	12.3	14.5	16.8	19.3	22.0
Surplus (+) / shortage (-)	-0.7	-1.3	-1.8	-3.9	-6.1	-8.6	-11.3
Spring and summer (April to September)							
Net generation	10.4	11.7	13.3	13.3	13.3	13.3	13.3
Transmission losses	1.4	1.6	1.3	1.3	1.2	1.1	1.1
Demand	7.8	9.3	11.1	13.1	15.2	17.5	20.0
Surplus (+) / shortage (-)	1.2	0.8	0.8	-1.1	-3.1	-5.3	-7.7

Source: Petromarket RG, based on the data provided by a national consultant

6.2. Additional power generation alternatives: price benchmarking

For the purpose of this study the following potential additional power generation options have been considered:

- Coal-fired TPP (power generation only) located near the Fon-Yaghnob coal deposit.
- Coal-fired TPP utilising supercritical steam conditions (“High Efficiency, Low Emissions” — *HELE*) located near the Fon-Yaghnob coal deposit (hereinafter, *HELE TPP*).
- Coal-fired TPP utilising supercritical steam conditions and CO₂ capture and storage, located near the Fon-Yaghnob coal deposit (hereinafter, *HELE+CCS TPP*).
- Coal-fired CHP (power and heat generation) located in Dushanbe.
- Gas-fired CHP located in Dushanbe.
- Solar power plant (hereinafter, *SPP*).
- Wind power plant (hereinafter, *WPP*).
- Rogun HPP completion project (additional investments of 4.2 billion USD are required to expand the plant from current 240 MW to planned 3 600 MW).
- HPP greenfield project (hereinafter, greenfield *HPP*).

To choose the most efficient option, alternatives were compared in terms of levelized cost of electricity (*LCOE*)¹¹. The technical and cost performance parameters of the potential power plants required for *LCOE* calculation (excluding the fuel costs) are detailed in **Table 6.3**.

¹¹ The levelized cost of electricity represents the average revenue per unit of electricity generated that would be required to recover the costs of building and operating a generating plant during an assumed financial life and duty cycle

Table 6.3 - Technical and cost performance parameters of potential power plants for year-round operation, 2022 conditions, 2021 prices

Type of plant	CAPEX (USD/kW)	OPEX (excluding the fuel costs), CAPEX percentage	Service life (years)	Capacity factor	Efficiency	Share of autumn and winter generation (October to March)
Coal-fired CHP	1 010 ¹	3.3% ²	30 ³	85% ²	27% ¹	50% ¹
Coal-fired HELE TPP	1 029 ^{1, 4}	3.3% ²	30 ³	85% ²	39% ^{1, 4}	50% ¹
Coal-fired HELE+CCS TPP	2 060 ^{1, 4, 5}	3.3% ²	30 ³	85% ²	39% ^{1, 4}	50% ¹
Coal-fired TPP	792 ^{1, 6}	3.3% ²	30 ³	85% ²	33% ²	50% ¹
Gas-fired CHP	758 ^{1, 2}	3.8% ²	30 ³	88% ²	50% ²	50% ¹
SPP	1 153 ³	3.0% ²	25 ³	18% ²	-	30% ⁷
WPP	2 092 ³	3.0% ²	25 ³	30% ²	-	50% ⁹
Rogun HPP	1 250 ^{10, 11}	1.3% ^{2, 12}	75 ²	41% ^{8, 2}	-	44% ¹⁰
Greenfield HPP	2 215 ²	1.3% ²	75 ²	41% ^{8, 2}	-	44% ¹⁰

Source: Petromarket RG, based on the references listed below the table

Notes:

1. Coal-fired Dushanbe CHP-2 performance data (with inflation-based correction of cost performance data, based on the data provided by a national consultant)
2. Tajikistan Power Sector Development Master Plan – Final Report, Asian Development Bank (in English), URL: https://mewr.tj/wp-content/uploads/files/Power_Sector_Master_Plan-Vol1.pdf
3. Projected Costs of Generating Electricity 2020 Edition, International Energy Agency (in English), URL: <https://iea.blob.core.windows.net/assets/ae17da3d-e8a5-4163-a3ec-2e6fb0b5677d/Projected-Costs-of-Generating-Electricity-2020.pdf>
4. Outlook and Benefits of an Efficient U.S. Coal Fleet, Wood Mackenzie (in English), URL: <https://nma.org/wp-content/uploads/2019/01/Outlook-and-Benefits-of-An-Efficient-U.S.-Coal-Fleet.pdf>; Cost–Benefit Analysis of HELE and Subcritical Coal-Fired Electricity Generation Technologies in Southeast Asia, Hassan Ali et.al (in English), URL: https://www.researchgate.net/publication/349000886_Cost-Benefit_Analysis_of_HELE_and_Subcritical_Coal-Fired_Electricity_Generation_Technologies_in_Southeast_Asia
5. Annual Technology Baseline 2018 z., National Renewable Energy Laboratory (in English), URL: <https://atb-archive.nrel.gov/electricity/2018/index.html?t=cc>
6. Combined Heat and Power, Energy Technology Systems Analysis Program International Energy Agency (in English), URL: https://iea-etsap.org/E-TechDS/PDF/E04-CHP-GS-gct_ADfinal.pdf
7. General Assessment of Global and Tajikistan Power Sector, Newsletter of the Academy of Sciences of the Republic of Tajikistan (in Russian), URL: <http://cawater-info.net/library/rus/petrov-ahmedov-kabutov-karimov-1.pdf>
8. Projections of GHG emissions to 2030 in Tajikistan, United Nations Development Programme (in English), URL: https://www.undp.org/sites/g/files/zskgke326/files/migration/tj/undp_tjk_Report_GHG_Projections_Tajikistan_2030.pdf
9. It is assumed that wind power generation does not depend on the season.
10. Data provided by a national consultant.
11. Remaining CAPEX, excluding the funds already spent.
12. Out of the total Rogun HPP project cost.

The following assumptions were used with regard to fuel prices¹²:

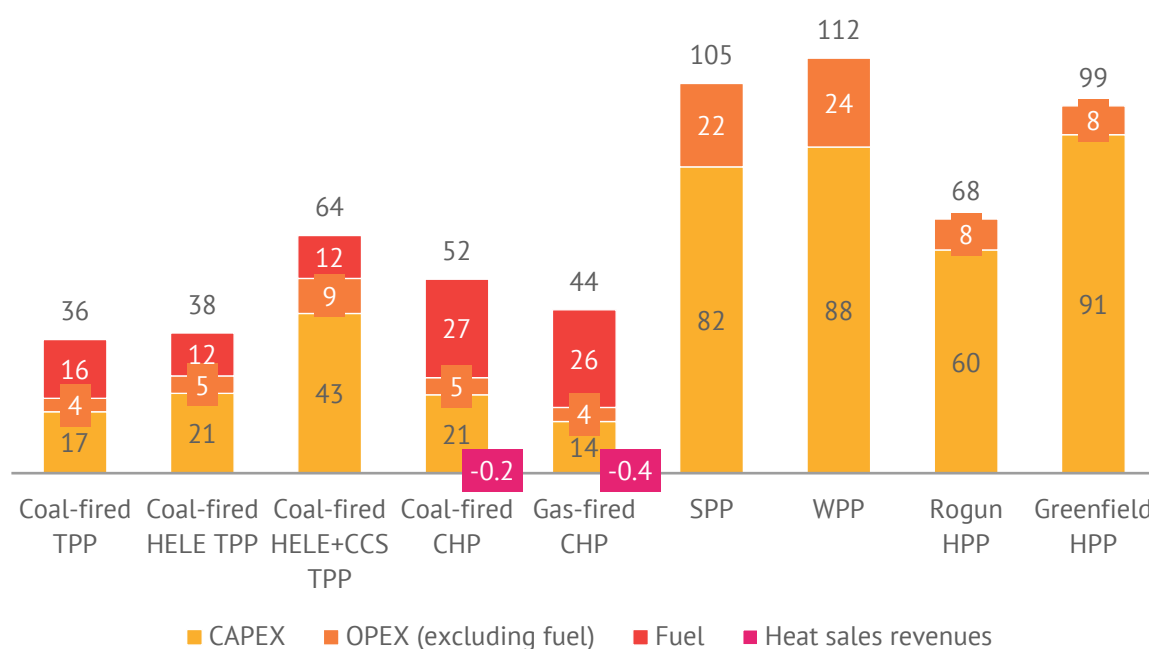
- For TPPs located near the Fon-Yaghnob coal deposit, the cost of coal was estimated at 300 somoni (27 USD) per tonne.
- For Dushanbe CHP, the cost of coal is the same, i.e. 300 somoni (27 USD) per tonne plus the cost of its delivery to Dushanbe estimated at 120 somoni (11 USD) per tonne.
- The cost of natural gas imported from Uzbekistan to CHP in Dushanbe was assumed to be 130 USD/thousand m³.

The cost of heat supplied by CHPs in Dushanbe was assumed to be 17 somoni (1.5 USD)/Gcal based on 2021 and 2022 tariffs, the price at which Dushanbe Heating Network Company buys heat from CHPs.

The real discount rate was assumed to equal 10%. This is a standard rate for developing countries (see IEA's study "*Projected Costs of Generating Electricity 2020 Edition*").

These assumptions are sufficient to calculate the levelized cost of generating 1 MW of power at various plants. The calculation results for all power plants based on their year-round operation are detailed in **Figure 6.1**.

Figure 6.1 - Levelized cost of electricity (LCOE) for potential power plants based on their year-round operation, USD/MWh in 2021 prices



Source: Calculations by Petromarket RG based on various sources (see Table 6.3)

Under the adopted assumptions, the cheapest power generation option is a conventional coal fired TPP ($LCOE = 36$ USD/MWh), and the cheapest “clean” option is completion of the Rogun HPP project, which is nevertheless 90% more expensive ($LCOE = 68$ USD/MWh).

¹² Data provided by a national consultant

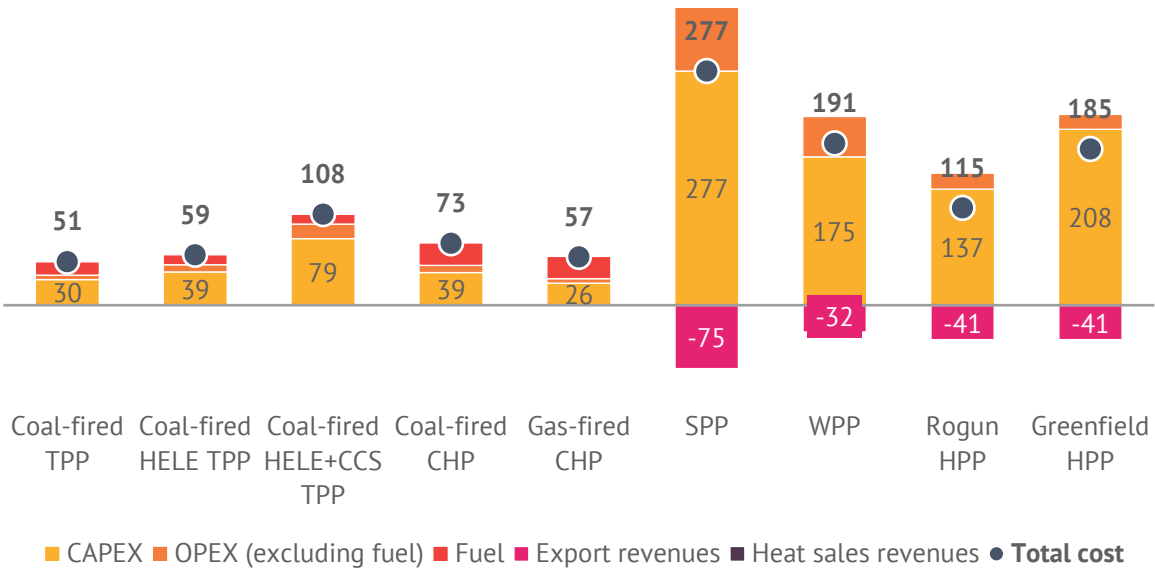
However, the requirement for additional power generation in spring and summer will emerge no earlier than in 2035 (see **Table 6.2**). Until then growing power demand in the domestic market during that period can be met by merely reducing electricity exports.

On the other hand, power shortages in autumn and winter exist already. This means that power plants commissioned before 2035 shall either operate in autumn and winter only (which is possible for coal-fired and gas-fired power plants) or export surplus electricity generated in spring and summer.

In this context, we disregard the issue of power grid throughput capacity for export power transmission (though this issue is currently relevant and causes the Nurek HPP, which is the largest HPP in Tajikistan, to sometimes switch to idle operation by discharging water without power generation). Upgrade and development of power grids constitute a separate task encompassing various projects implemented in Tajikistan, including those sponsored by international development partners.

Figure 6.2 demonstrates the comparison of levelized cost of electricity for the autumn and winter-oriented option. In this option, TPP and CHP are on generation duty in autumn and winter only, while SPP, WPP and HPP are in year-round operation and sell surplus electricity generated in spring and summer for export at a price of 32 USD/MWh (2022 average weighted selling price of electricity for export according to the national consultant).

Figure 6.2 - Levelized cost of electricity (LCOE) in autumn and winter for potential power plants, USD/MWh in 2021 prices



Source: Calculations by Petromarket RG based on various sources (see Table 6.3)

As indicated by **Figure 6.1** and **Figure 6.2**, the levelized cost of electricity for season-oriented generation is higher than for year-round duty. For TPP and CHP, this is dictated by the fact that the same CAPEX value is distributed across the smaller amount of annually generated power in MWh. For SPP, WPP, and HPP, it can be explained by the fact that the export price of electricity is below the levelized cost of electricity at such plants. Hence, power generation at SPP, WPP, and HPP in spring and summer does not pay off, and to compensate the costs, it is necessary not only to export “surplus” electricity, but also to increase the cost of electricity for domestic users in autumn and winter.

In case of season-oriented generation, increase in the levelized cost of electricity makes no difference in terms of the result: a coal-fired TPP remains the cheapest option ($LCOE = 51$ USD/MWh), and its nearest competitor among the “clean” energy, which is Rogun HPP, turns out to be more than twice as expensive ($LCOE = 115$ USD/MWh).

However, selection for power plant construction cannot be based only on a simple arithmetic comparison of levelized cost of electricity. Consideration shall also be given to such points as public health (Sub-Section 6.3) and climate (Sub-Section 6.4) costs associated with power plant operations. Moreover, potential funding sources for construction of various power plants (Sub-Section 6.5) are to be considered.

Before giving attention to these issues and comparing a coal-fire TPP (as the most competitive option) with clean generation alternatives (primarily, Rogun HPP completion, with greenfield HPP to be built afterwards), the following features of other types of plants are also worth noting, as they are also to be kept in mind when selecting the best option for additional power generation:

- **A coal-fired HELE TPP** utilising supercritical steam conditions yields quite a low LCOE, but in Tajikistan’s conditions is still more expensive than a conventional coal-fired TPP, as due to comparatively low local coal prices and high discount rate, the increased CAPEX cannot be compensated by reduced OPEX. If, however, coal prices were to rise or a carbon emission fee or lower discount rate were introduced, HELE TPP could become more competitive than a conventional coal fired TPP.
- **Construction of a HELE+CCS TPP** in Tajikistan is deemed not feasible, primarily in terms of economics, as there is no explicit carbon emissions pricing (which could pay off the increased cost of generation). In addition, utilizing the carbon capture technology at TPP is not practical, as the country lacks not only the CO₂ transportation and storage infrastructure, but also any prerequisites for its development. The world’s most common CO₂ storage option, namely re-injection into oil reservoirs to enhance oil recovery, cannot be applied in Tajikistan as there is virtually no oil production in the country (less than 25 thousand tonnes in 2021).
- **A gas-fired CHP** is competitive as compared to a coal-fired TPP, which is partially due to cheap gas imported from Uzbekistan. Tajikistan and Uzbekistan cooperate under the existing informal agreement stipulating that Tajikistan exports electricity to Uzbekistan at a low price (20 USD/MWh), and, in return, receives natural gas at a reduced price (130 USD/thousand m³). Agreements such as this one bears a certain obvious risk. Should the political situation between Tajikistan and Uzbekistan change, there would be a risk of a substantial increase of the gas price. At the same time, no such risk exists with regard to coal prices, as they are dictated by the mining cost borne by the national producers.
- In this context, it is worth remembering that there was already an instance of a gas supply disruption, which took place in 2012. Despite the fact that the relations with Uzbekistan have significantly improved since then, construction of additional natural gas fired power plants would increase Tajikistan’s dependency on its neighbour.
- Once the **Rogun HPP** is commissioned, its water reservoir will become the key regulator for the Vakhsh River water discharge (together with the current key regulator, i.e. the Nurek water reservoir). In the past, Uzbekistan voiced concerns about the Rogun HPP, which it feared would cause a shortage of irrigation water for its land. These concerns effectively blocked the possibility of project funding by the development partners of Tajikistan. However, these concerns have been alleviated, and therefore the project no longer faces barriers to seeking international funds (Sub-Section 6.5).

- A **greenfield HPP** is not assumed to have its own water reservoir capable of inter-seasonal regulation of water discharge, but since it will be situated downstream of the Rogun HPP on the Vakhsh River, its seasonal generation profile is likely to be identical to that of Rogun HPP.

6.3. Effect of coal-based generation on public health

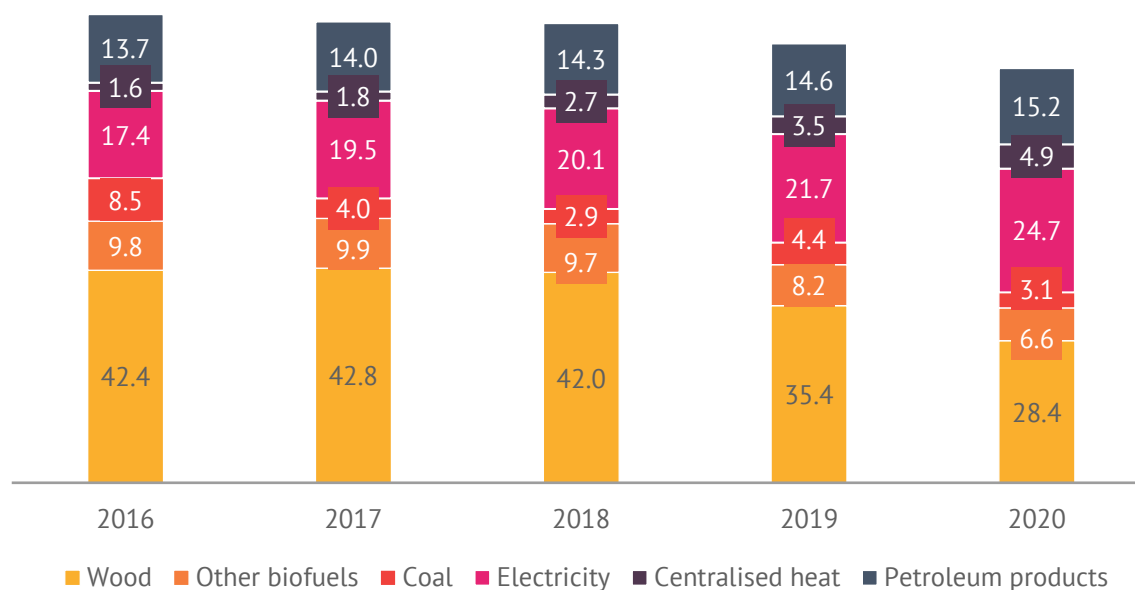
As the effects caused by operation of additional coal-fired power plants (and associated additional coal extraction) on health of the citizens of Tajikistan is assessed, two separate options shall be considered: (1) utilizing coal-based generation to eliminate the existing power shortage, and (2) utilizing coal-based generation to meet the growing power demand in future.

In case of the power shortage elimination, the adverse health impact from increasing coal-based generation will be by far compensated by reduction of emissions of PM2.5 particulate pollutants resulting from combustion of solid biofuels and coal by the households.

In Tajikistan there is a high share of deaths caused by household air pollution. According to the World Bank's study *"The Global Health Cost of PM2.5 Air Pollution: A Case for Action Beyond 2021"*¹³, 35% of deaths caused by air pollution in Tajikistan (2 607 out of 7 365 deaths from air pollution in 2019) are related to household air pollution resulting from burning wood, other biofuels, and coal.

This results from the power consumption profile of Tajikistan's households (see **Figure 6.3**). Due to power shortage, to meet various domestic needs, including space heating in autumn and winter, the population uses primarily wood and other solid biofuels (manure, stalks of maize, and cotton plants), as well as coal and petroleum products.

Figure 6.3 - **Gross power consumption by population for domestic needs in Tajikistan, actual data for 2016, estimated data for 2017 to 2020, TJ**



Source: *Enquiry on the status of power facilities and efficiency of their utilisation in 2016, calculations by Petromarket RG*

¹³ Text of the document in English, URL: <https://openknowledge.worldbank.org/bitstream/handle/10986/36501/9781464818165.pdf?sequence=4&isAllowed=y>

Even if it is assumed that the observed 2017-2020 growth in the population's consumption of power and heat from the centralized heat supply sources is fully a result of the people's shift away from solid biofuels, the latter still holds a substantial share in the gross power consumption of the country.

At this point, it is worth noting that even though as gross power consumption declines, the net power consumption is estimated to increase at the population growth rate. Such difference in gross and net consumption trends is associated with low thermal efficiency of burning wood, other biofuels, and coal in home stoves. The estimate is based on the assumption that thermal efficiency is 35% for burning coal and 25% for other biofuels (as no similar data are available for Tajikistan, the quoted numbers are taken from the World Bank's study on Kyrgyzstan, namely "*Fueling Kyrgyzstan's Transition to Clean Household Heating Solutions*"¹⁴).

Eliminating the existing power shortage will discourage some households from using "dirty" fuels, which, according to estimates made by the authors of this study, will reduce Tajikistan's death rate from air pollution indoors from 2,607 to 1,852 deaths per year. It should be emphasised that the demand for wood, other biofuels, and coal, and, consequently, the death rate from air pollution indoors, will not decrease to zero. A lower demand for "dirty" fuels and a reduced death rate will match the substitution of solid biofuels and coal for 1 billion kWh of power consumption (the existing level of shortage). Some households which currently stock up on wood and other biofuels by collecting the fuel themselves at no charge (as purchased wood makes up only a third of the overall consumption of that fuel by the population), are not likely to give up such fuels in exchange for electricity, which would come at a certain price.

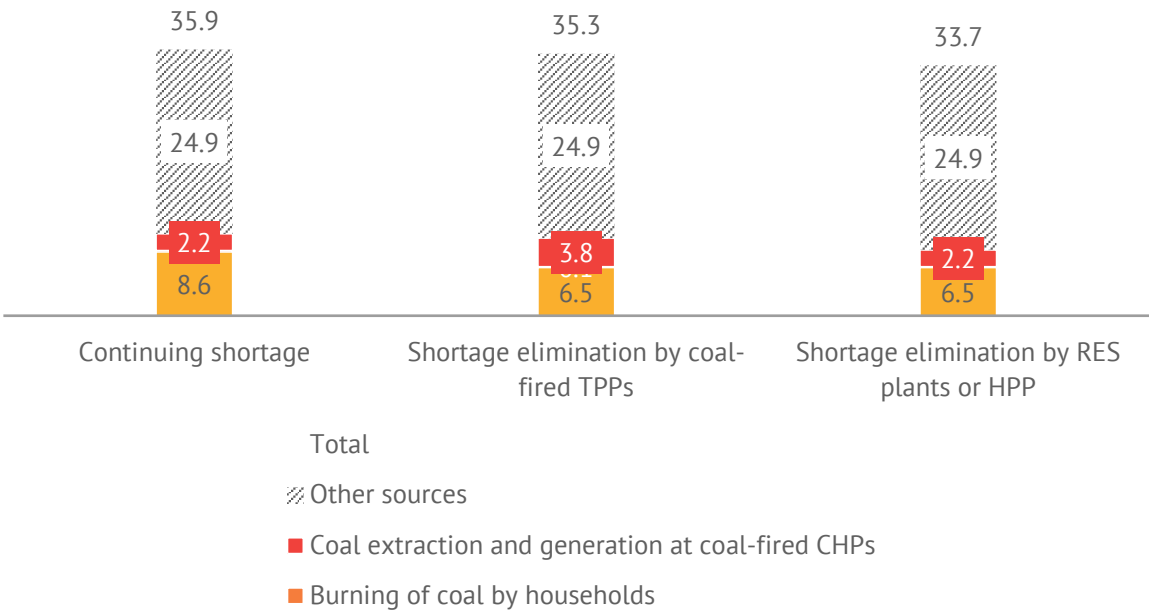
It is worth noting that there is no difference in the results of reducing the death rate from household air pollution, regardless of whether the electricity is provided by coal-fired or RES power plants. It is because in both cases certain households will obtain access to electricity and therefore discontinue burning solid biofuels and coal on their own. However, the difference between coal-fired and RES power plants will still exist in terms of their input into ambient air pollution, as expansion of coal-fired power plants will result in greater PM2.5 particulate emissions from power generation and coal extraction facilities. Nevertheless, even taking that factor under consideration there will be still a positive net effect from eliminating the power shortage by means of coal-based generation (as emissions will be reduced; see **Figure 6.4**).

Households burning solid biofuels and coal are not only the source of household air pollution but make also an input equal to 24% into ambient air pollution¹⁵. As a result, eliminating the power shortage by providing coal-based energy will, on the one hand, lead to greater emissions from coal extraction and generation at coal-fired TPPs, and, on the other hand, help reduce air pollution by households. At the same time, eliminating the power shortage by clean generation will result in an even greater emission reduction, since alike TPPs, it will allow to eliminate certain amount of the emissions from the households, and, contrary to TPPs, it will not account for any additional emissions from power generation.

¹⁴ Text of the document in English, URL: <https://openknowledge.worldbank.org/bitstream/handle/10986/33833/Fueling-Kyrgyzstan-s-Transition-to-Clean-Household-Heating-Solutions.pdf?sequence=1&isAllowed=y>

¹⁵ Text of the document in English, URL: <https://www.nature.com/articles/s41467-021-23853-y#MOESM4>

Figure 6.4 - Contribution of various sources into ambient air pollution, μg of PM2.5 particulate matter/cubic m



Source: McDuffie, E.E., Martin, R.V., Spadaro, J.V. et al. Source sector and fuel contributions to ambient PM2.5 and attributable mortality across multiple spatial scales, calculations by Petromarket RG

Moreover, when coal-fired power plants are considered as a means not a as a solution to cover the existing shortages, but to meet the growing power demand for the future, their construction does not give any positive effect in terms of the reduced wood, other biofuels, and coal consumption by the households.

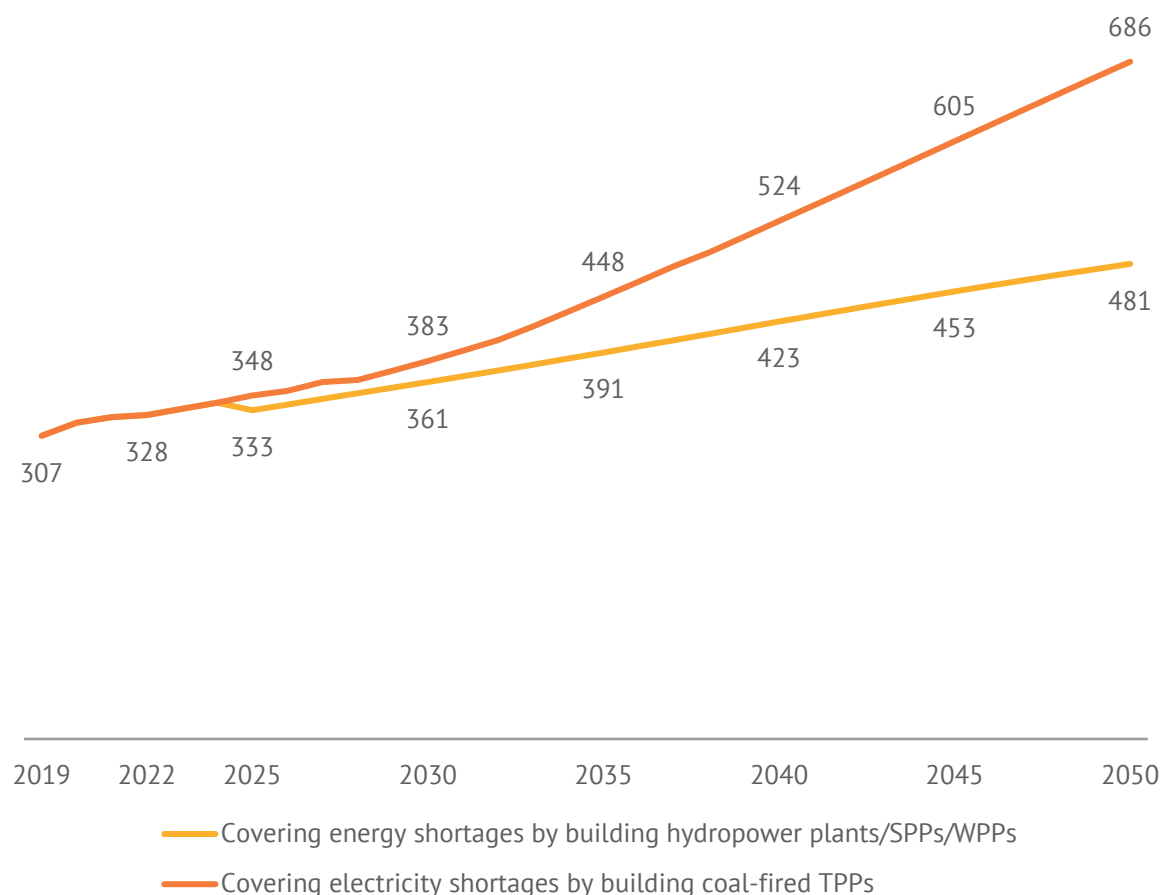
To assess the adverse impact of coal-based generation on public health, health damage from air pollution (measured in monetary terms) was forecasted under the assumption that all sources of pollution apart from coal extraction and coal-based generation will keep their emissions at fixed levels (see **Figure 6.5**).

It was assumed that the level of emissions from coal-fired TPPs would change along with the growing coal-based generation capacities required to meet the increasing power demand. The same figure demonstrates also the forecasted health damage trends if the emerging power demand is to be met by means of HPPs and RES power plants (with the emissions from coal fired TPPs fixed at the existing level).

The fact that health damage increases even when the additional power demand is covered by RES or HPPs can be explained by the fact of the population growth, as greater population results in a higher amount of deaths at the same level of air pollution.

As it can be seen from **Figure 6.5**, the difference between coal-based and clean power generation in terms of annual public health damage by 2050 will amount in monetary form to 205 million USD. Per 1 MWh of additional generation, the damage is estimated at about 11 USD/MWh, which is a relevant amount, but still many times less that the difference in LCOE between a coal fired TPP and Rogun HPP.

Figure 6.5 - Expenditure associated with health damage from ambient air pollution for the growing power demand to be met by coal-fired TPPs vs. RES power plants (assuming that contributions of other emission sources will remain unchanged once the power shortage is eliminated), actual data for 2019, estimated data for 2020-2021, forecast data for 2022-2050, million USD/year



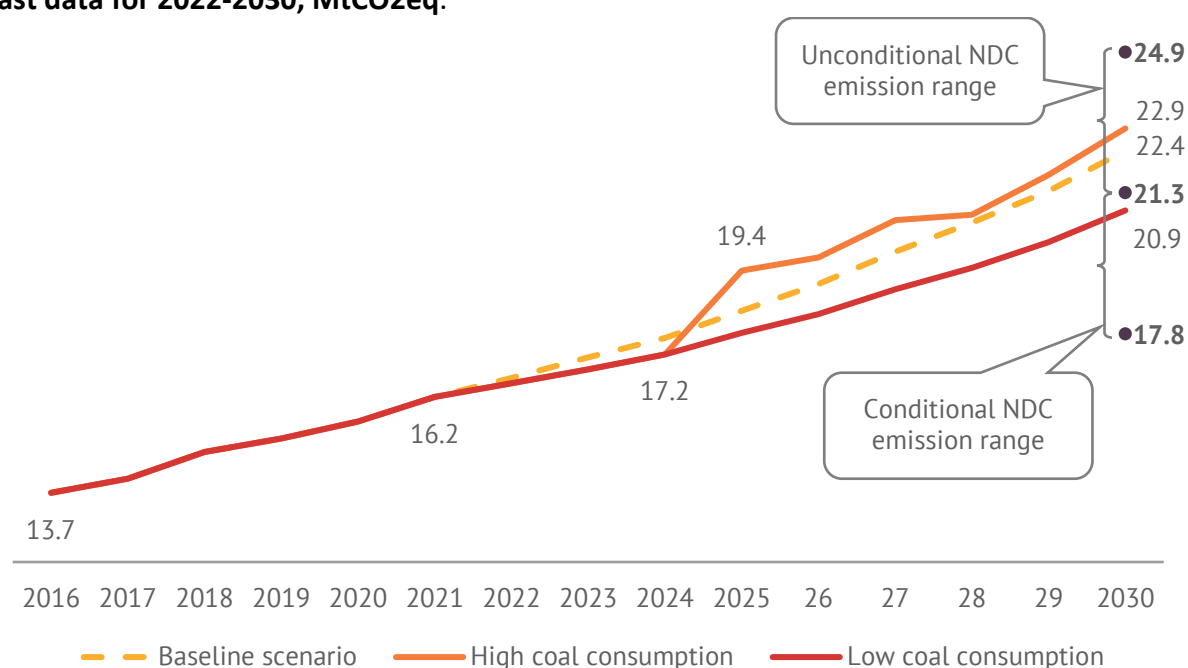
Source: *The Global Health Cost of PM2.5 Air Pollution: A Case for Action Beyond 2021*, calculations by Petromarket RG

6.4. Climate concerns of coal-based generation

The Republic of Tajikistan does not employ any explicit pricing mechanisms applicable to the emissions of CO₂ and other greenhouse gases (hereinafter, GHG), and no such mechanisms are expected to be adopted in the foreseeable future. Under these circumstances CO₂ emissions originating from coal-fired power plant operations are not reflected in coal-based generation costs.

In the baseline scenario assuming that no efforts will be made to reduce emissions in any sector of the Tajik economy and that the existing fuel profile will remain unchanged in the power and heat generation sector, between 2021 and 2030 the emissions from the power sector will increase by 1.5 MtCO₂eq. per year, and the overall GHG emissions will grow by 6.2 million tonnes to 22.4 million tonnes per year, taking into account the absorbing capacity of forests (see **Figure 6.6**).

Figure 6.6 - Greenhouse gas emissions forecast for Tajikistan until 2030 in the baseline (the power sector structure remains the same as in 2016, no additional efforts are made to cut emissions), high (the additional power requirement is fully met by coal-fired TPPs), and low (the requirement is covered by RES or HPPs) coal demand scenarios, actual data for 2016, estimated data for 2017-2021, forecast data for 2022-2030, MtCO₂eq.



Source: Calculations by Petromarket RG based on the Republic of Tajikistan Fourth National Message under the UN Framework Convention on Climate Change and the Study on Greenhouse Gas Emissions Forecast for Key Sectors and Assessment of Effect from Climate Change Mitigation Policy and Efforts. Greenhouse Gas Emissions Forecast for the Republic of Tajikistan until 2030, updated version of the Republic of Tajikistan nationally determined contribution (NDC). Note: Due to varying data on greenhouse gas emissions in different official publications, this calculation was made by Petromarket RG independently based on actual GHG emission data taken from the Republic of Tajikistan Fourth National Communication, and based on the emission growth rates in various sectors within the forecast period taken from the UNDP prescribed Study on Greenhouse Gas Emissions Forecast for Key Sectors and Assessment of Effect from Climate Change Mitigation Policy and Efforts. Greenhouse Gas Emissions Forecast for the Republic of Tajikistan until 2030

In the scenario assuming a high coal consumption by the power generation sector, in which the additional power demand not covered by the ongoing projects (see **Table 6.1**) is fully met by coal-fired TPPs (being the cheapest power generation option), generation-specific emissions will grow by 2 MtCO₂eq. per year by 2030, which is only 500 ktCO₂eq. more per year than in the baseline scenario. The increase in fugitive emissions from coal extraction is 27 ktCO₂eq. per year. In this scenario, the cumulative net GHG emissions in 2030 will amount to 22.9 MtCO₂eq. per year, which is within the country's established 2030 unconditional NDC target of 21.32 to 24.87 MtCO₂eq. per year.

In the context of the above, it is worth noting that the emission data from coal mining in Tajikistan are estimated using emission factors, and the coefficients for methane emissions from surface mining are 12.5 times less than those for underground mining.

Assuming that all additional coal is extracted from underground mines, an increase of the emissions from mining would then amount to 340 ktCO₂eq. per year (27 ktCO₂eq. per year estimated for surface mines).

If the additional power demand is fully met by RES power plants or HPPs, by 2030 the GHG emissions from the power sector will not grow above the 2021 level, and the cumulative emissions (amounting to 20.9 MtCO₂eq. per year) will fall within the conditional target range.

Tajikistan's capability to meet its unconditional target of GHG emissions reduction by 2030 even in the case of using only a coal-based generation to cover the additional power demand results from the following:

1. relatively unambitious targets, as detailed in the country's NDC (the unconditional target will be achieved even if the GHG emissions grow by as much as 80% as compared to 2016, which is the last year for which the actual data is available), and
2. expected low demand for additional generation (above the already proposed projects) by 2030.

However, fully covering the additional power requirement by 2050 using only coal fired TPPs will result in the increase in GHG emissions by 19.8 MtCO₂eq per year from the power generation sector only, which is by itself almost 1.5 times more than the cumulative emissions from all sectors in 2016.

Such increase in emissions will impede the country from achieving its carbon neutrality goal, which at the moment has not been officially set and is discussed in the country only as a concept. Such goals have already been declared by all other former USSR Central Asian states except Turkmenistan (Kazakhstan — by 2060, Kyrgyzstan and Uzbekistan — by 2050), as well as by the Tajikistan's main regional trade partners (Russia and China — by 2060).

Moreover, in the long-term scenario, the Carbon Border Adjustment Mechanism (CBAM) might be introduced not only in the EU states, where the time horizon for its establishment is expected to be relatively short, but also in other trading partners of Tajikistan, such as e.g. in China. If indirect GHG emissions (i.e. emissions from generation of electricity used in manufacturing CBAM-covered products) were included in the calculation of the carbon footprint of products subject to CBAM, a decision to pursue low-carbon rather than fossil-based power generation will result in increased competitiveness for Tajikistan's goods in importing states' markets. Moreover, if most other countries in the UNECE region invest in "clean" generation, giving preference to development of coal-based generation in Tajikistan would deprive the country from an opportunity to capitalize on its current advantage, which it enjoys due to a high share of hydropower in its cumulative power generation.

Given a long service life of coal-fired power plants (approx. 30 years) and trends in international climate policy (increasing international pressure on countries to commit to climate neutrality and promotion of CBAM mechanisms), current investments into coal-based generation may entail challenging and expensive future efforts to reduce GHG emissions and maintain competitiveness. Therefore, such investment is very likely not to pay off due to the highly probable necessity of early decommissioning of the newly constructed coal-fired power plants.

6.5. Funding of power generation projects: obstacles and opportunities

In recent years, a large number of countries and institutional banks declared that they would cease to fund projects in the coal-fired power industry abroad. For instance:

- In November 2021 at COP26, 39 countries and institutional banks signed the statement on international support for the clean energy transition,¹⁶ which includes the commitment to cease funding of unabated power projects associated with fossil fuels. Among the signatories are e.g. USA, UK, Canada, and the European Investment Bank.

¹⁶ Text of the document in English, URL: <https://ukcop26.org/statement-on-international-public-support-for-the-clean-energy-transition/>

- In April 2021 at the Leaders Summit on Climate, South Korea declared that it would no longer provide state support for coal-fired projects abroad.¹⁷
- In May 2021, Japan together with the other G7 participants committed to stopping state support for unabated coal-fired project abroad.¹⁸
- In September 2021 at the UN General Assembly, China committed to discontinuing construction of new coal-fired power plants abroad.¹⁹

Out of all the above examples, the last one is the most important for Tajikistan. In our study, low CAPEX (as compared to construction costs in the Western countries), which gives a competitive advantage to coal-fired power plants in Tajikistan, is achieved precisely by using Chinese relatively inexpensive technologies in local power plant construction projects.

For instance, Dushanbe CHP-2 was built by TBEA, a Chinese company, and 95% funded from a subsidized loan provided by Exim Bank, a Chinese institutional bank (with the remaining 5% funded by the state-owned power holding OSHC "Barki Tojik").

OSHC "Barki Tojik" is unable to obtain a loan from any private bank due to its high debt burden. According to the 2020 audit statements²⁰, the total debt of the company amounted, as of 31 December 2020, to 32.6 billion somoni (2.9 billion USD) vs. only 3.3 billion somoni (0.3 billion USD) 2020 gross annual revenue and 6.6 billion somoni (0.6 billion USD) net losses before tax.

The remaining option for funding coal-fired power plants in Tajikistan is to do so from the national budget. The cost of constructing a 300 MW coal-fired power plant (which would be sufficient to eliminate the country's power shortage in winter) near the Fon-Yaghnob coal deposit will be as low as 238 million USD, while the 2022 national budget revenue was established at 2,920 million USD²¹ (at an exchange rate of 11.3 somoni/USD).

However, 415 million USD of the budget revenues are international grants allocated to implement public investment projects. Additional grants could probably be raised for RES power plants and HPPs, but not for coal-fired power plants, as the above statements on stopping the funding of coal-fired power plants were always accompanied by the statements on increasing the funding for RES plants. This logic holds true also in practice: the approved RES projects indicated in **Table 6.1** are implemented with the financial support of the World Bank, USAID, and Asian Development Bank. In addition, according to Reuters,²² in July 2022, the European Commission contacted the European Investment Bank with a request to take part in the funding of the Rogun HPP project and became its "largest investor".

As OSHC "Barki Tojik" is very limited in terms of independent funding for a coal-fired power plant construction project, the decisions of international development partners become a crucial factor in the selection of a future power plant type for Tajikistan. **This means that no new coal fired power plants will be built in Tajikistan.**

¹⁷ Text of the document in English, URL: <https://www.whitehouse.gov/briefing-room/statements-releases/2021/04/23/leaders-summit-on-climate-summary-of-proceedings/>

¹⁸ Text of the document in English, URL: <https://www.gov.uk/government/publications/g7-climate-and-environment-ministers-meeting-may-2021-communiqué/g7-climate-and-environment-ministers-communiqué-london-21-may-2021>

¹⁹ Text of the document in English, URL: <https://news.un.org/en/story/2021/09/1100642>

²⁰ Text of the document in English, URL: http://barqitojik.tj/upload/iblock/102/FS_%20Barqi%20Tojik_eng_%202020_G_1607.pdf

²¹ Text of the document in Russian, URL: <http://ncz.tj/content/закон-республики-таджикистан-о-государственном-бюджете-республики-таджикистан-на-2022-год>

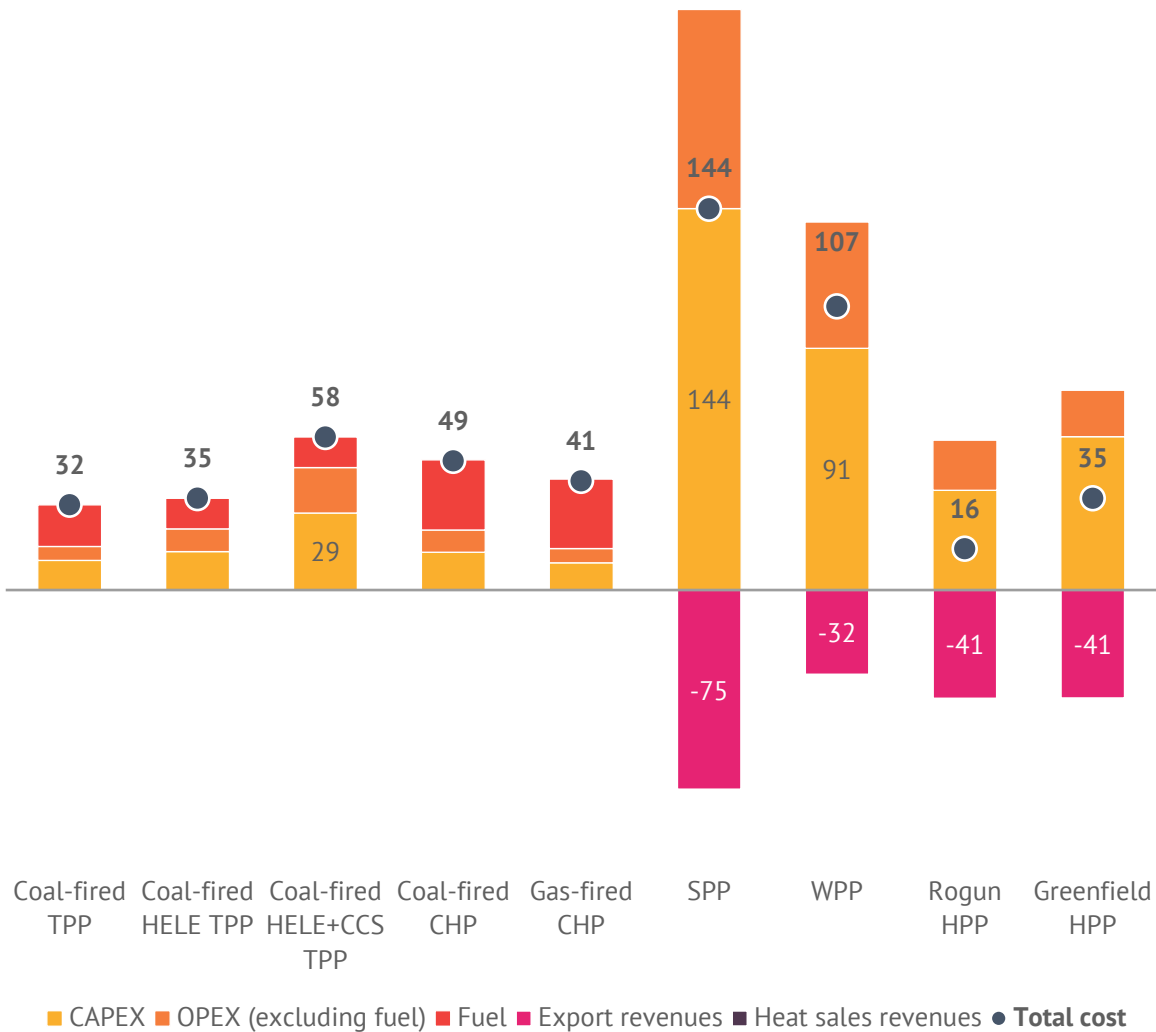
²² Text of the document in English, URL: <https://www.reuters.com/article/ukraine-crisis-eu-tajikistan-idAFL8N2YM2FR>

It is important to note that the fact that the international development partners have ceased funding of coal-fired power plants, does not mean that they decided to ignore the seemingly most cost competitive option of reducing power shortage in the country only for environmental reasons.

When selecting projects for financial support, institutional organisations apply a much lower discount rate, which reflects the value of various investments for the public (the so-called social discount rate is herein assumed at 3%), but not in terms of pay-off.

If the social discount rate of 3% (see **Figure 6.7**) is used to compare different types of power plants, then Rogun HPP completion would provide additional power generation in autumn and winter at a cost half as high as the cost of building a new coal-fired power plant. Generation by the new HPP is only 3 USD/MWh more expensive than generation by a new coal-fired power plant, and if health effects due to air pollution by coal-based generation are taken into account, it will be 8 USD/MWh cheaper.

Figure 6.7 - Levelized cost of electricity (LCOE) in autumn and winter for future power plants at a discount rate of 3%, USD/MWh in 2021 prices



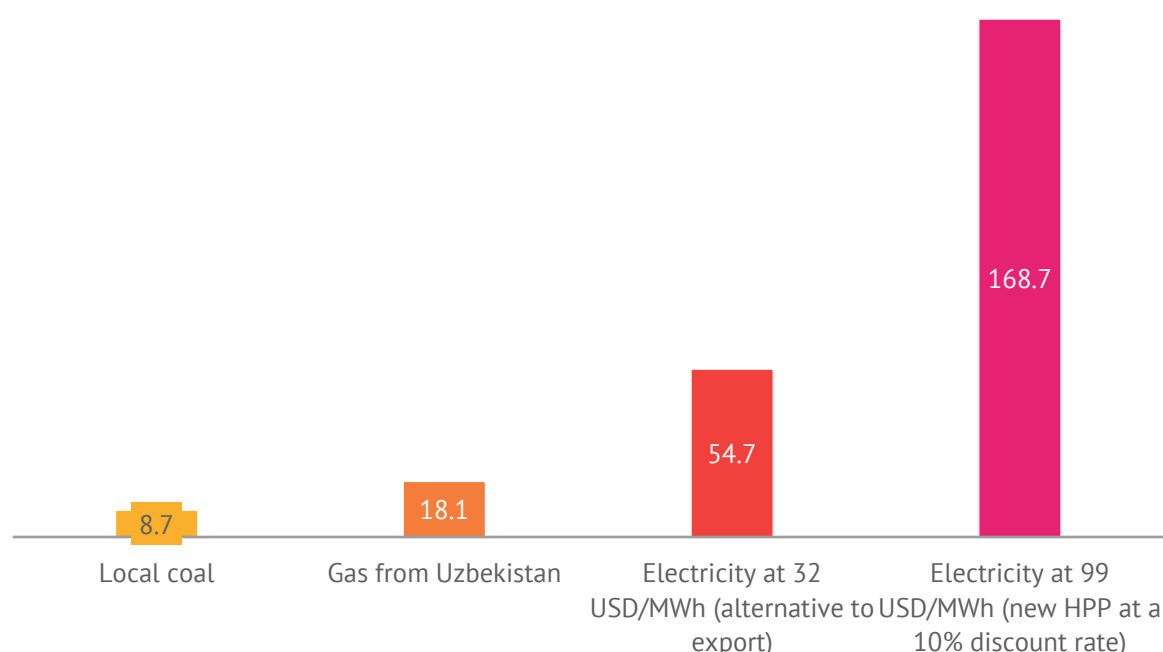
Source: Calculations by Petromarket RG based on various sources (see Table 6.3)

7. Coal demand forecast in Tajikistan until 2050

7.1. Coal demand by industrial consumers in Tajikistan

As mentioned in Section 4, more than 90% of coal demand by Tajikistan's industrial facilities are currently attributed to the local cement industry. Coal demand by facilities in other sectors started to decline after Uzbekistan resumed provision of natural gas in 2018, as gas is for most industries a more cost-effective fuel option than coal. That is not the case however, in the cement industry for which coal turns out to be by far the most cost-effective option (see **Figure 7.1**).

Figure 7.1 - **Cost of fuel for cement production using various fuel types, USD/tonne of cement**

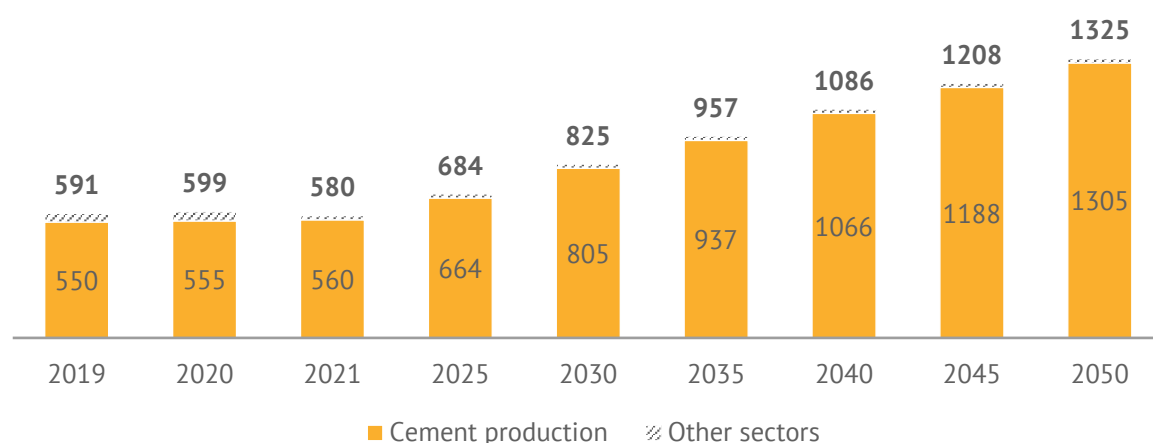


Source: Calculations by Petromarket RG

Natural gas doubles the cost of fuel for cement plants, while electricity increases such costs more than 6-fold, even if it is provided at a relatively low price, at which it is exported in spring and summer (32 USD/MWh). Based on the estimated cost of electricity at a new HPP, the ratio approaches a 20-fold value. Such a high cost for the electricity option is due to the specific cement manufacturing process, which requires high temperatures that can be reached by means of either plasma torches or by burning hydrogen electrolysed from water. Both ways are extremely energy intensive.

It is therefore reasonable to assume that further industrialization of Tajikistan, in all sectors apart from cement, will rely on natural gas rather than coal. For that reason, the forecast of coal demand by the entire pool of industrial consumers (see **Figure 7.2**) is based on the assumption that the future demand for coal will grow only in the cement sector, whilst in all other industries in the country it will remain at the level from 2021.

Figure 7.2 - Coal demand by industrial facilities, including cement facilities, actual data for 2019-2021, forecast data for 2022-2050, thousand tonnes



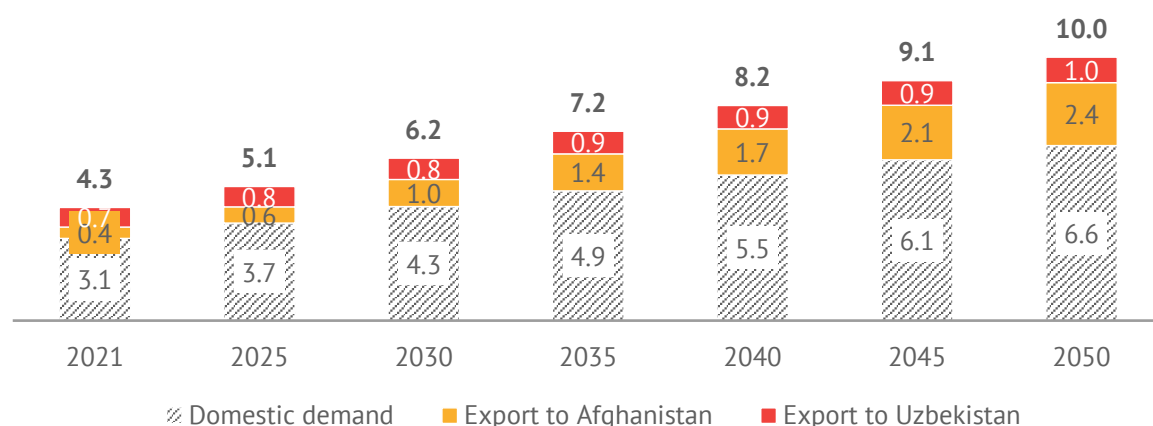
Source: Calculations by Petromarket RG

The forecast of coal demand by the cement industry was, in turn, based on the cement production forecast for Tajikistan (see **Figure 7.3**), which relies upon the assumptions that: (1) the demand for cement by the domestic market will be fully covered by local producers, and (2) the share of Tajikistan's products in satisfying the demand for cement in Uzbekistan and Afghanistan (which are the main importers of cement from Tajikistan) will remain unchanged during the forecast period.

When estimating the cement demand in Tajikistan, Uzbekistan, and Afghanistan, consideration was given to their expected populations and welfare growth rates (as cement consumption per capita tends to increase with the improvement of living standard).

A more intensive growth of cement production through emergence of new export routes is unlikely, as cement has a high weight/price ratio. This means that the future competitiveness of Tajikistan's cement on external markets will depend on availability of cheap means of transportation, which in Tajikistan's case, given its geography, means availability of rail. For the moment, Tajikistan has a railroad connection only with Uzbekistan. Any development of the national network allowing for connections with other neighbours is in the short- and mid-term perspective unlikely.

Figure 7.3 - Cement production in Tajikistan, actual data for 2021, forecast data for 2022-2050, million tonnes



Source: Calculations by Petromarket RG

7.2. Demand for coal as non-energy feedstock

Apart from using coal as a fuel, the industry may demonstrate the demand for coal as feedstock for production of e.g. coke (needed by the iron and steel industry), synthetic gas or synthetic petroleum products, or petrochemical products.

While coking coal is being extracted in Tajikistan (from a number of formations at the Fon-Yaghnob deposit), it is currently used solely as fuel. Its use for other purposes would be possible if there were iron and steel industry facilities in the country. However, at the moment that it is not the case, despite the fact that in 2016 the Tajik Steel Mill was commissioned in Gissar. The company was allocated in the area of the Fon-Yaghnob coal deposit and managed to obtain a reduced electricity fare. At the same time, it failed to secure supplies of iron ore raw materials and therefore had to cease its operations in 2018.

As the circumstances have neither changed, nor are they likely to change in any foreseeable future, it is quite implausible that a demand for coking coal to supply Tajikistan's domestic iron and steel industry will emerge in a short- or a mid-term perspective. The fact that the global steel industry has started to gradually limit the use of coking coal and test the feasibility of switching to alternative technologies is not a decisive factor in this case. What is instead, is the lack of any apparent prospects for the development of the iron and steel industry in the country. Presently, there are simply no convincing answers to a number of key questions on that matter, such as: which actual feedstock sources could the future local iron and steel industry rely upon, would local products be competitive to the imported ones, is the demand on the domestic market sufficient to gain the required effect of scale, what is the export potential of the iron and steel products, etc. The search for answers to the above questions goes beyond the scope of this study. However, considering a very telling example of the described-above failed commissioning of the local steel mill, for the purpose of the analysis of the future coal demand by Tajikistan's iron and steel industry, this forecast assumes that it will be null.

Tajikistan has experience in operating coal-to-gas conversion units. Such units were commissioned at the TALCO's aluminium smelter and the Dushanbe East Boiler House in 2012 after the natural gas supplies from Uzbekistan had stopped. However, once the latter were resumed in 2018, the units were decommissioned, as synthetic gas was not price competitive with the imported natural gas.

From Tajikistan's energy security point of view, however, it might be reasonable to install coal-to-synthetic gas conversion units in the country, so that they could be used in case of any future disruptions of natural gas supplies from Uzbekistan. However, with an unrestricted access to natural gas, as it is now the case, it makes no sense to utilize synthetic gas, as that would only undermine the competitiveness of the local products, in particular, the aluminium produced by TALCO. The forecast scenario does not anticipate that natural gas supplies from Uzbekistan will stop again, and therefore estimates that the future demand for coal to produce synthetic gas in Tajikistan would be zero.

Coal-to-petroleum conversion is likely to be as uncompetitive as the production of synthetic gas. Consequently, the products relying on energy supplies generated from the obtained petrol will be more expensive than those imported from Russia (a more accurate estimate of synthetic petroleum product cost is beyond the scope of this study). Reliance on such products would therefore entail additional expenditures for the population and business.

In addition, the demand for petroleum products in Tajikistan is expected to decrease as the country intends to vigorously promote electric vehicles on the domestic market. In October 2022, the country adopted the *Electric Transport Development Program for 2023-2028*, while the *Republic of Tajikistan's Green Economy Development Strategy for 2023-2037* set the goal to increase the share of electric vehicles to 55% by 2037.

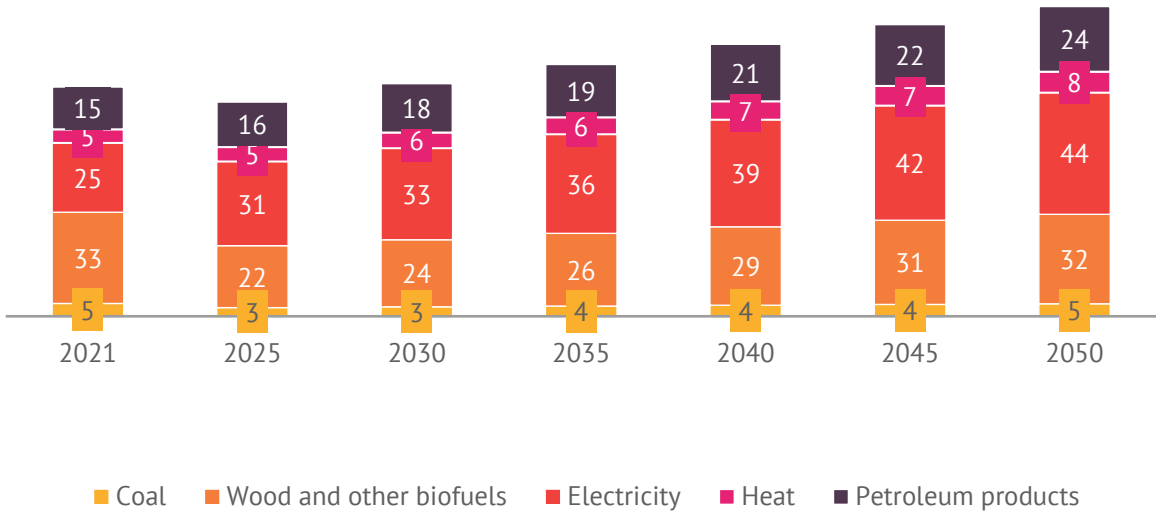
Under such circumstances, investments into coal-to-petroleum conversion facilities seems unreasonable, and therefore, also in the case of this sector, the forecast of coal demand is zero.

Coal-to-chemicals processes based on the coal-to-olefins technology could technically be more feasible from the economic point of view. The issues of product price and competitiveness, sufficient capacity of the domestic market and feasibility to start such plant are, however, beyond the scope of this review. As Tajikistan does not have such projects (even at the feasibility study stage), the forecast does not take into account the coal demand by petrochemical facilities.

7.3. Coal demand by households in Tajikistan

By 2025 the yearly coal demand by households will decrease from 273 thousand tonnes recorded in 2021 to 185 thousand tonnes. Then it will increase again to 268 thousand tonnes by 2050. In other words, by mid-century it will almost return to its 2021 level. Such u-shaped trend in coal demand is due to the anticipated completion of Rogun HPP. The existing obstacles to the project funding are expected to be removed, and the power plant is to be ramped up to its full capacity of 3 600 MW in 2033. It is assumed that the funds for that purpose will be allocated in the near future, and that the resulting gradual increase of the Rogun HPP's capacity will allow to eliminate the current power shortage by 2025. The shortage affects the population mainly in autumn and winter, and as soon as it is eliminated, households are expected to reduce their consumption of not only wood and other solid fuels, but also coal (see **Figure 7.4**).

Figure 7.4 - Gross power consumption by households in Tajikistan, estimated data for 2021, forecast data for 2022-2050, TJ



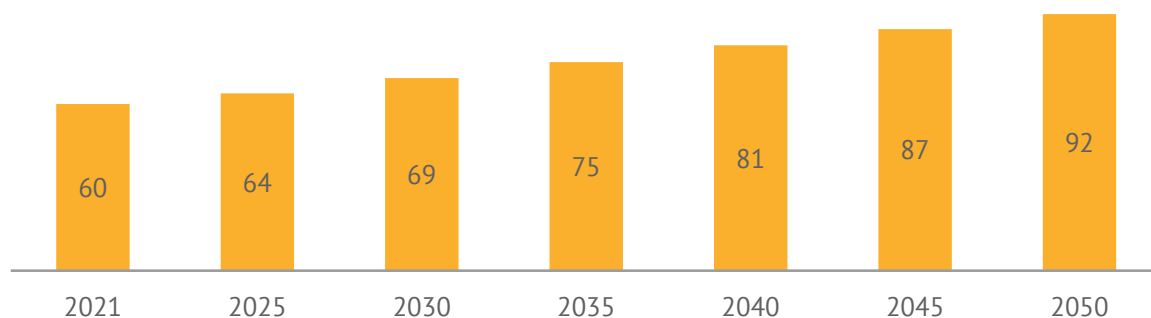
Source: Enquiry on the status of power facilities and efficiency of their utilisation in 2016, calculations by Petromarket RG

No changes are expected in the fuel consumption profile by the population after 2025, i.e. it is assumed that the coal demand will grow in proportion to the corresponding increase in energy requirements. The forecasted rates of growth in the energy demand for the heating purposes by the population are assumed equal to the population growth rates. The expected growth in cumulative energy demand per capita resulting from the increase of income per capita will be primarily covered by electricity and will not affect coal consumption.

7.4. Coal demand by public organisations in Tajikistan

The demand for coal by public organisations (such as schools, hospitals, as well as security, defence, and law enforcement agencies) will increase within the forecast period in proportion to the growth in the number of public organisations, which will, in turn, correspond to the population growth (see **Figure 7.5**).

Figure 7.5 - Coal demand by public organisations, actual data for 2021, forecast data for 2022-2050, thousand tonnes



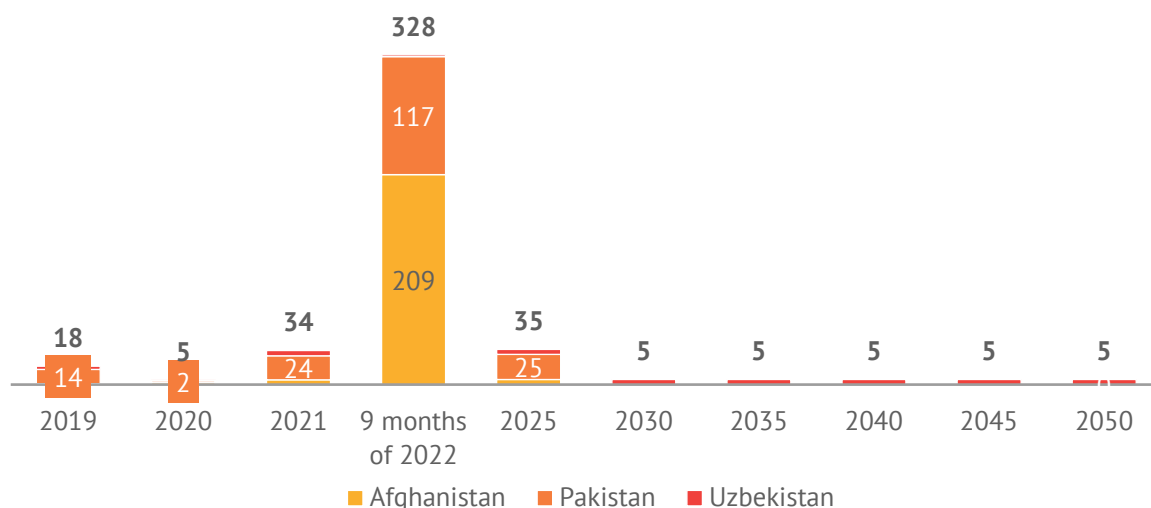
Source: Data provided by a national consultant, calculations by Petromarket RG

Technically, the demand for coal by the public organisations could also include the needs of the future coal-fired boiler houses constructed for the provision of the centralised heating. However, the centralised heating supply system is currently in operation only in Dushanbe, where the proposed new boiler houses are expected to run on natural gas rather than on coal.

7.5. Export potential of Tajikistan coal

Despite the soaring coal export from Tajikistan in 2022 (328 thousand tonnes of coal exported in 9 months of 2022 vs. a total of 34 thousand tonnes in 2021), there are no long-term prospects for Tajikistan coal export in the future (see **Figure 7.6**).

Figure 7.6 - Coal export from Tajikistan, broken down by countries, actual data for 2019-2022, forecast data for 2023-2050, thousand tonnes



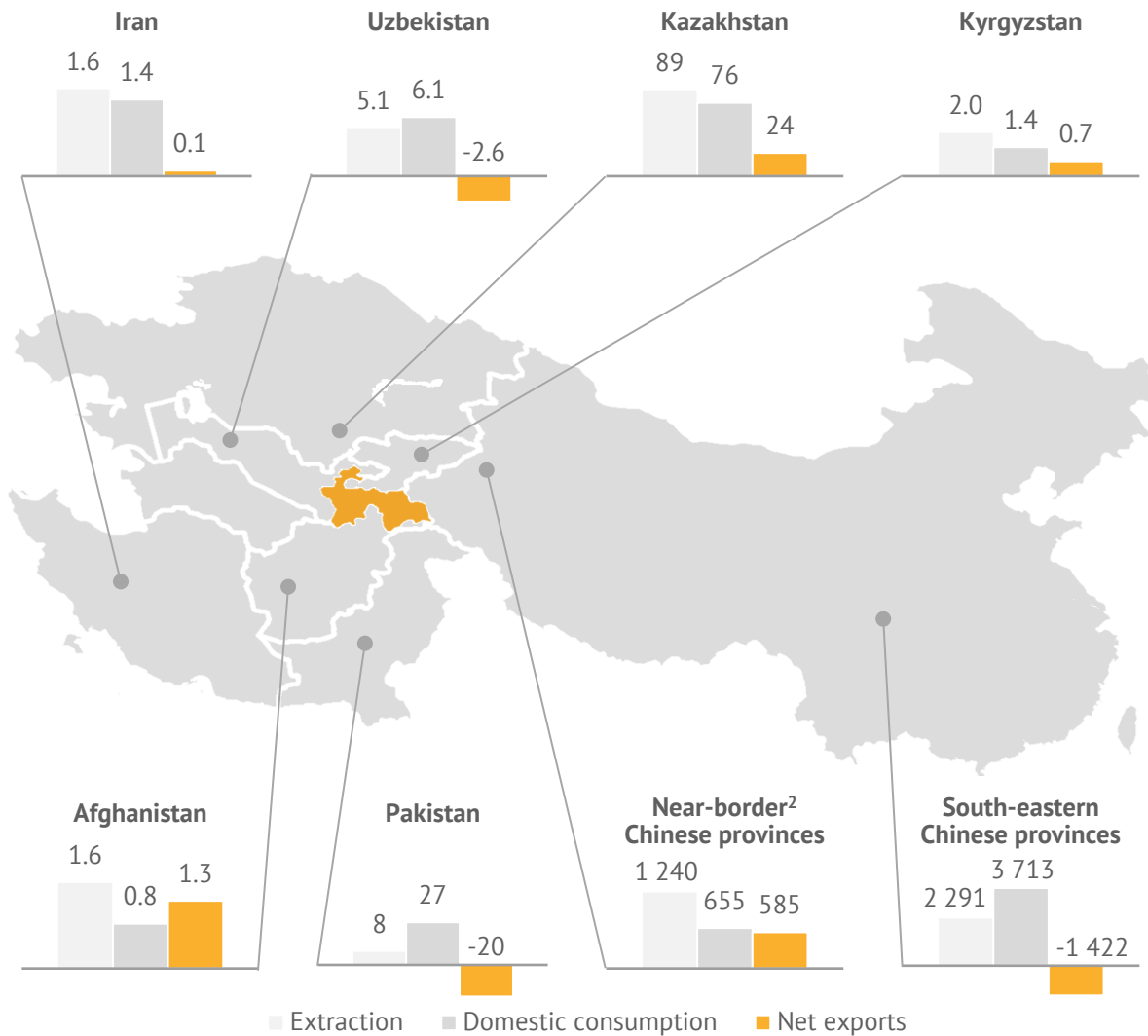
Source: Data provided by a national consultant, calculations by Petromarket RG

Most neighbouring countries of Tajikistan are net coal exporters (see **Figure 7.7**). The two exceptions are Uzbekistan and Pakistan. Despite generally being a net coal importer, China cannot be treated as a suitable export market for the Tajikistan's coal since the provinces located in proximity to the Tajikistan's borders (i.e. those to which Tajikistan's coal could be technically delivered by road) act collectively as net coal exporters.

Exceptionally high 2022 export to Afghanistan results from the fact that Afghan companies exported the locally extracted coal to Pakistan, covering the domestic needs with the imports from Tajikistan.

The fast growth of export to both Afghanistan and Pakistan in 2022 was due to the significant increase in the global coal prices caused by the geopolitical upheaval. Such situation made deliveries of the cheaper Tajikistan's coal (see Sub-Section 3.3) competitive despite their high road transportation costs, which under normal conditions render such purchases uneconomical (Tajikistan coal deposits are not connected to the railway network, neither there is a direct railroad service between Tajikistan and Afghanistan or Pakistan).

Figure 7.7 - Coal extraction, consumption and net export in the neighbouring countries of Tajikistan, 2018 data for China, 2021 data for other countries, million tonnes



Source: U.S. Energy Information Administration, UN Comtrade, calculations by Petromarket RG

Notes:

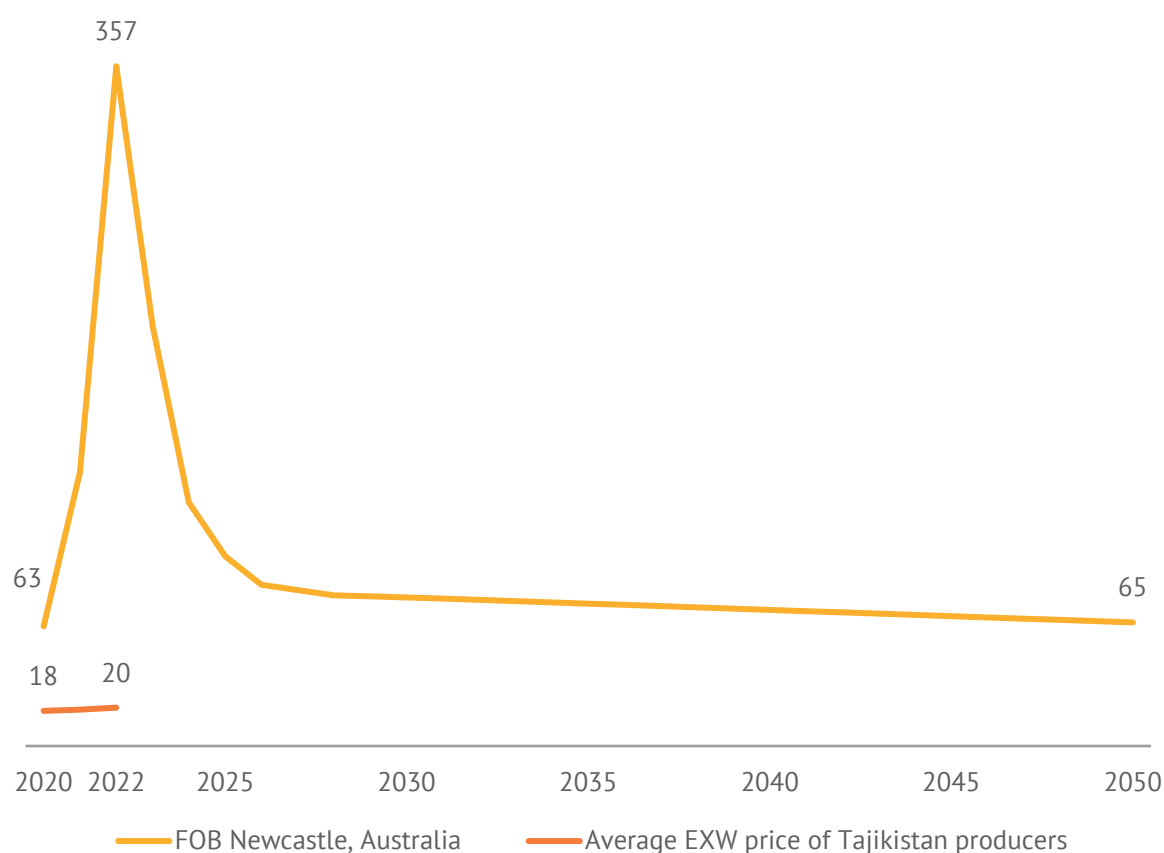
1. Extraction, domestic consumption and net exports of coal in Turkmenistan are virtually absent

2. Near-border Chinese provinces are Gansu, Inner Mongolia, XUAR, Qinghai, and Tibet

When the global coal prices are at 60-70 USD/tonne, Tajikistan coal is unable to compete on the Pakistani market with the coal delivered there from other countries by the sea. Based on the review of the actual coal export contracts, the cost of coal delivery from the Tajikistan's Fon-Yaghnob deposit to Islamabad (Pakistan) amounts to at least 80 USD/tonne. However, since in 2022 the global coal prices increased almost six-fold as compared to 2020, the coal deliveries by road from Tajikistan became competitive.

Given that in the forecast period, coal prices are very likely to drop back to the 2020 level, the export to Pakistan and Afghanistan will slump down by 2025 and become null by 2030.

Figure 7.8 - Power-generating coal prices on FOB Newcastle basis and average EXW prices of Tajikistan producers, actual data: 2020-2022, forecast data: 2023-2050, USD/tonne



Source: IMF Primary Commodity Price System, calculations by Petromarket RG

It is technically possible to increase coal export to Uzbekistan, which is much closer to the main coal deposits of Tajikistan than Pakistan. However, for that to happen the price needs to be competitive with the prices of Kazakhstan's and Kyrgyzstan's producers who deliver coal to Uzbekistan by rail. Logistics costs of the Tajikistan's coal deliveries can be reduced by connecting the deposits of Tajikistan (primarily Fon-Yaghnob) to the railway network, but in view of the goal to achieve carbon neutrality declared by Uzbekistan, an infrastructural project such as this is risky and may not pay off. Hence, this study assumes the forecast export to Uzbekistan to remain at the existing low level (of approx. 5 thousand tonnes per year).

7.6. General insight into Tajikistan coal market

The review discussed in Sections 6 and 7 concludes that cumulative demand for Tajikistan coal will be only 1.5 times higher by 2050 than in 2021 (see **Table 7.1**).

Demand by the power sector will not change during the forecast period: Dushanbe CHP-2 will keep operating as usual, i.e. in autumn and winter only (its potential replacement with alternative power plants is discussed below in Section 8), but no new coal-fired power plants will be built.

The most intensive growth will be demonstrated by the manufacturing industry (almost 2.5 times) due to the growing coal requirement by the cement industry.

Table 7.1 - Demand for Tajikistan coal by domestic and foreign consumers, actual data for 2020-2021, forecast data for 2022-2050, thousand tonnes

Parameter	Actual data		Forecast data					
	2020	2021	2025	2030	2035	2040	2045	2050
Domestic demand	2,043	1,813	1,994	2,172	2,341	2,507	2,666	2,816
Power sector	1,206	900	991	991	991	991	991	991
Manufacturing industry	599	580	755	911	1,058	1,200	1,336	1,465
State-funded organisations	57	60	64	69	75	81	87	92
Households	181	273	185	201	218	235	252	268
Export	5	34	35	5	5	5	5	5
Total	2,047	1,847	2,029	2,177	2,346	2,512	2,671	2,821

Source: Calculations by Petromarket RG

The cumulative coal demand (less than 70 million tonnes in total in the period of 2022-2050) will be fully met by the local producers and will not require additional exploration, as the current commercial coal reserves in Tajikistan are estimated at 317.2 million tonnes.

8. Prospects of coal replacement with alternative local energy sources

As indicated by the forecast presented in the previous section, the coal demand reduction may potentially occur in the power sector, provided that the Dushanbe CHP-2 plant is decommissioned, and in household sector, if the power shortage is eliminated. At the same time, in all other segments, the presented forecast does not envisage a possibility for coal demand reduction.

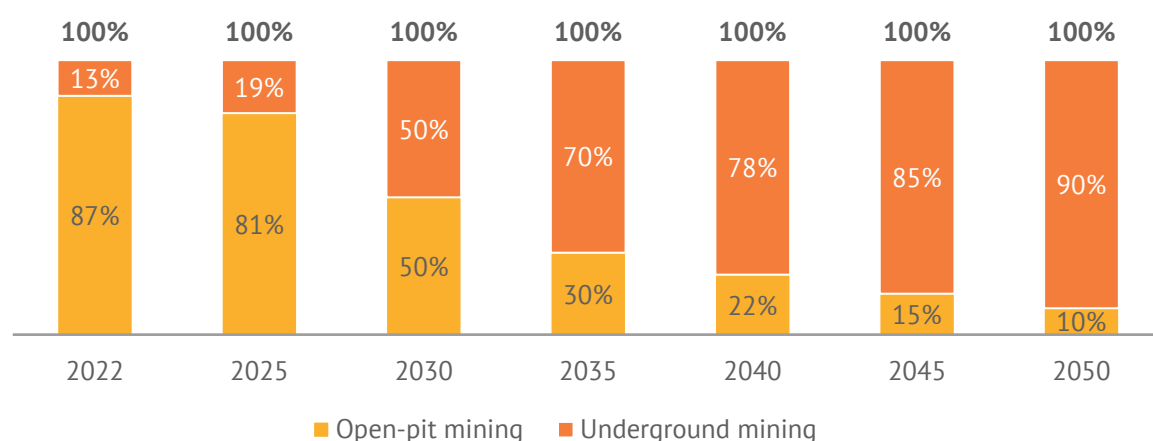
- The demand for coal by industrial consumers comes primarily from the cement facilities, for which coal is the only competitive fuel due to its low price (see **Figure 7.1**).
- The demand for coal by public organizations will not reduce either, since for this to happen, the organizations currently relying on their individual coal-based heating systems would need to get connected to either the centralized heat supply, or a gas pipeline, which in both cases is possible only in a limited number of regions of Tajikistan.
- As noted in Sub-Section 7.3, the coal demand by population will decrease in a short-term once the power shortage is eliminated. However, the further power supply growth (above what is necessary to cover the shortage) is unlikely to lead to further reduction of consumption of coal and solid biofuels by population, as electrical heat is more expensive than the heat derived from aforementioned fuels, and a portion of the population will continue to use those despite the associated health damage.

Considering the above, to answer the question whether the coal demand in Tajikistan will decrease during the forecast period, it is necessary to evaluate whether at any point in the future the cost of electricity generation by a potential new non-coal-fired power plant is to be lower than the cost of electricity production at the existing coal-fired CHP.

The cost of generation in the future will change. While the cost of coal-based generation will grow due to increasing coal prices and the maintenance costs of the aging existing plants, the solar- and wind-based generation will gradually become cheaper as with time those technologies will develop and mature.

The coal price growth will result from a steady decrease of the local extraction in open-pit mines and the simultaneous growing reliance on the underground mining (see **Figure 8.1**). Such change will occur because 80% of the opportunities for open-pit coal mining in Tajikistan are already exhausted.

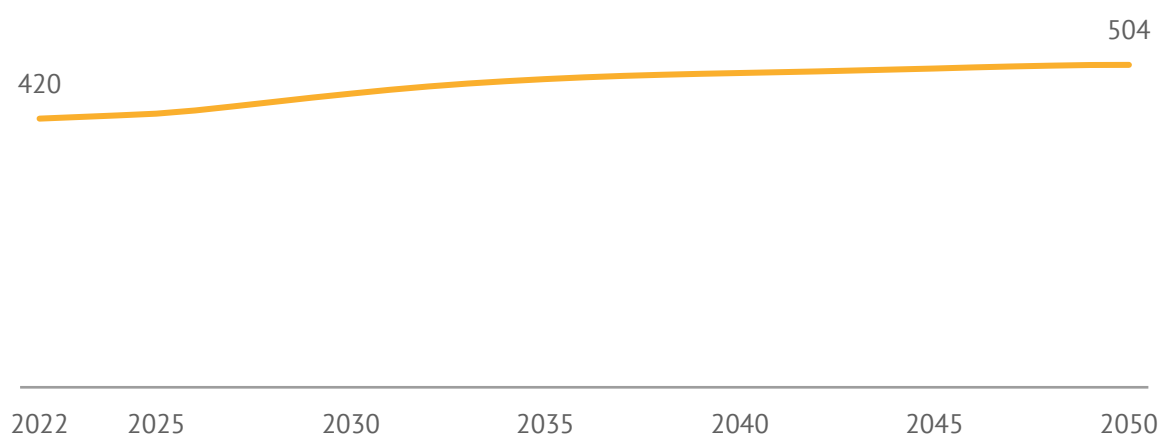
Figure 8.1 - Coal open-pit mining vs. underground mining in Tajikistan, actual data for 2022, forecast data for 2022-2050



Source: Petromarket RG, according to the data provided by SUE Angishti Tojik

Given the conditions in Tajikistan, underground coal mining is 40% more expensive than the open-pit method. Therefore, it is estimated that the coal price calculated on the basis of the Dushanbe CHP-2 costs of purchase (i.e. the price including the cost of delivery from the deposit at 120 somoni (11 USD)/tonne) will be higher by 20% in 2050 as compared to its 2022 level (see **Figure 8.2**).

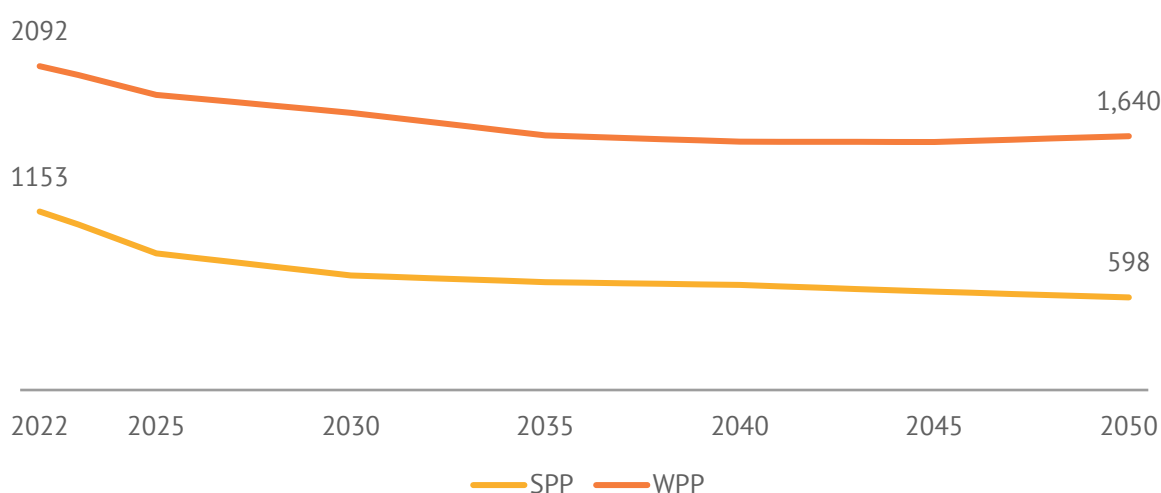
Figure 8.2 - Coal price including delivery to Dushanbe CHP-2, actual data for 2022, forecast data for 2023-2050, somoni/tonne in 2021 prices



Source: Petromarket RG, according to the data provided by SUE Angishti Tojik

However, as the number of the completed worldwide solar and wind power plants increases, their CAPEX (see **Figure 8.3**) and accordingly also OPEX (calculated as % of CAPEX), as well as the LCOE will decrease.

Figure 8.3 - CAPEX for solar and wind power plants, actual data for 2021, forecast data for 2022-2050, USD/kWh in 2021 prices



Source: Petromarket RG based on the following reports: "Renewable Energy Market Update - May 2022"²³ IEA and «bp Energy Outlook. 2022 edition»²⁴ bp

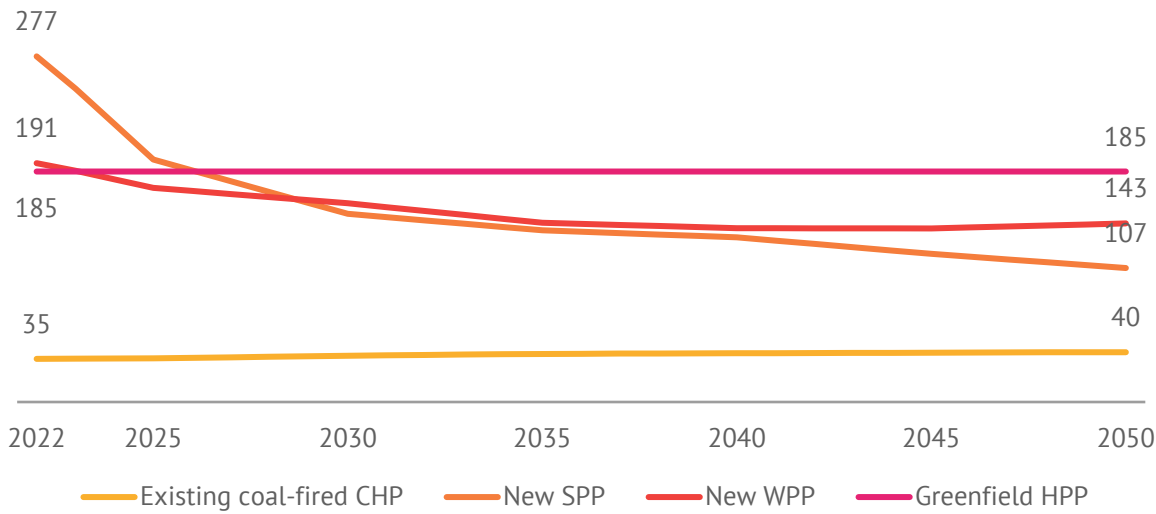
²³ Text of the document in English, URL: <https://www.iea.org/reports/renewable-energy-market-update-may-2022>

²⁴ Text of the document in English, URL: <https://www.bp.com/content/dam/bp/business-sites/en/global/corporate/pdfs/energy-economics/energy-outlook/bp-energy-outlook-2022.pdf>

As demonstrated by the graph above , despite the growth of coal prices and the decline of the costs of generation at SPP and WPP, if the funds for development of future generation capacities are raised without engaging international development partners (at a discount rate of 10%), coal-fired CHP remains the most cost effective option throughout the forecast period (see **Figure 8.4**).

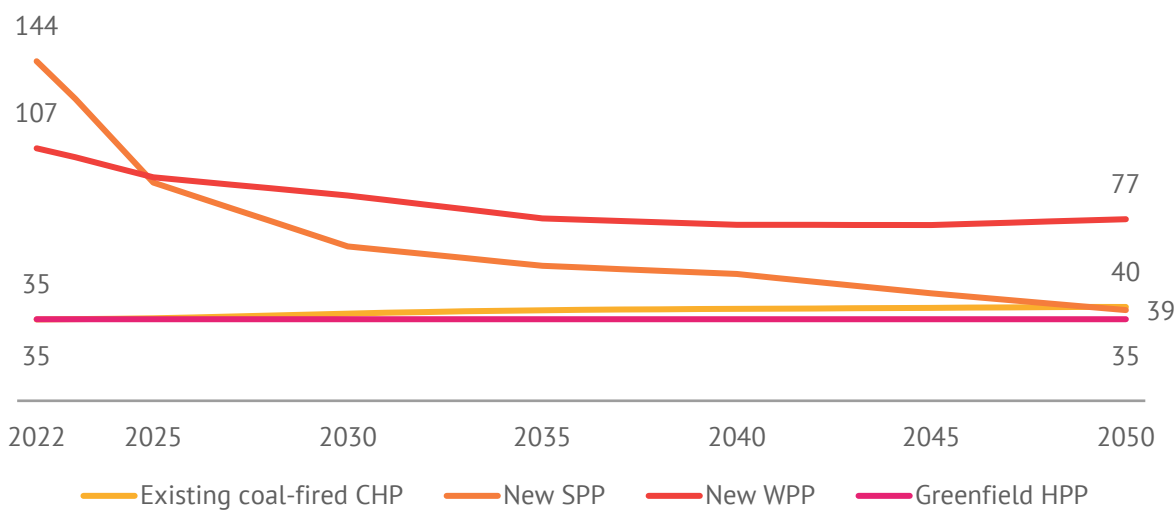
At the same time, if in the calculation a discount rate of 3% is utilized, which is the estimated rate if that funding is allocated by international development partners, then decommissioning of Dushanbe CHP-2 and its replacement with a new HPP would be cost effective even now (especially in view of health damage caused by the coal-fired power plant), and substitution with SPP will become cost effective approximately in 2050 (see **Figure 8.5**).

Figure 8.4 - Levelized cost of electricity (LCOE) in autumn and winter for the operating coal-fired Dushanbe CHP-2 vs. new solar, wind, and hydro power plants at a discount rate of 10% in 2022-2050, USD/MWh in 2021 prices



Source: Calculations by Petromarket RG

Figure 8.5 - Levelized cost of electricity (LCOE) in autumn and winter for operating coal-fired Dushanbe CHP-2 vs. new solar, wind, and hydro power plants at a discount rate of 3% in 2022-2050, USD/MWh in 2021 prices



Source: Calculations by Petromarket RG

In other words, if at present the international development partners were ready to allocate funds for constructing “clean” generating capacities sufficient to cover the current power shortage and substitute for the energy nowadays provided by Dushanbe CHP-2, then decommissioning of the latter could be considered. However, the fact that so far, no funds are allocated to cover at least the existing power deficiencies testifies against making an assumption that “clean” power generation will be funded to a greater extent in the foreseeable future. This means that CHP-2 will keep operating in autumn and winter.

The efforts made to mitigate power losses and introduce energy-saving technologies (see Sub-Section 2.1) may reduce the need for additional generation capacities, but their cumulative effect in autumn and winter (savings of 0.86 billion kWh by 2030) will be insufficient either to cover the existing power shortage (1 billion kWh) or to substitute for the existing CHP-2 output (1.32 billion kWh).

The seasonal peak in power demand in autumn and winter could be lowered by a widespread use of heat pumps that are 2 or more times more efficient than conventional electric heaters (i.e. have more than 200% efficiency). However, their initial cost is too high for both, households with moderate, or lower income, as well as for the State (supported by the international development partners), which does not have sufficient means to execute a campaign for their mass application.

9. Conclusions and recommendations

The period of the intensive growth of coal demand in Tajikistan occurred in 2012 to 2018, when Uzbekistan discontinued its natural gas supplies to the country. During that period, the coal-fired Dushanbe CHP-2 plant was built in an accelerated mode, and a number of industrial facilities shifted from natural gas to coal (used either directly as fuel, or as coal-to-synthetic gas feedstock). As a result, over that time the coal demand demonstrated a 7-fold growth from 253 thousand tonnes in 2011 to 1 741 thousand tonnes in 2018.

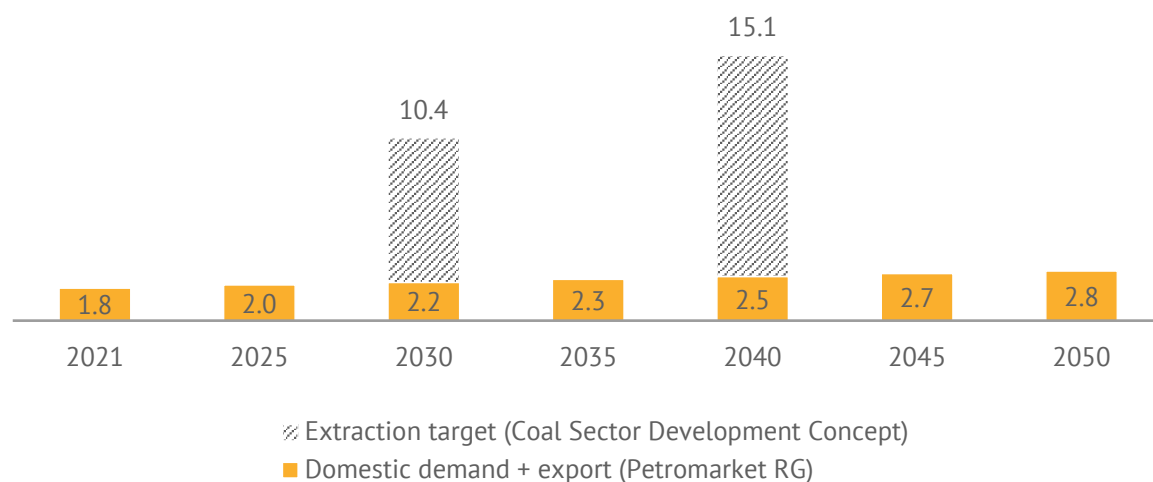
It was during that period, *the Republic of Tajikistan's National Development Strategy until 2030* was adopted. Based on the condition governing then, it naturally assumed that the coal sector would grow in the future. The extraction was expected to grow to 10.4 million tonnes by 2030 in the more conservative scenario, and to 15.1 million tonnes in the more optimistic one.

The Coal Sector Development Concept adopted in 2019 confirmed the target for coal extraction of 10.4 million tonnes in 2030 and set it for 15 million tonnes in 2040.

However, since 2019, the coal demand growth decelerated drastically. The yearly average demand in 2019-2021 was about 2 million tonnes, even though in accordance with the *National Strategy*, coal extraction in 2020 was supposed to reach between 4.1 and 5.3 million tonnes.

Such breaking point in the coal demand trend occurred primarily due to the changes in the political environment. The natural gas supplies from Uzbekistan were resumed, which encouraged industrial facilities in most sectors (except the cement industry) to return to using natural gas as a fuel of choice. At the same time, the prospects of the coal-fired power industry growth have been undermined because, among other things, they did not fit the country's climate agenda. In addition, the international development partners, also driven by the climate change considerations, committed to stop funding projects for unabated coal-based power generation. Finally, China, which had previously contributed to the development and execution of the Dushanbe CHP-2 construction project, also made a commitment not to fund any further coal-fired power projects abroad.

Figure 9.1 - Demand for Tajikistan coal (domestic market + export) as predicted in the current study, and the target coal extraction volumes according to the Coal Sector Development Concept, actual data for 2021, forecast data for 2022-2050, million tonnes



Source: Petromarket RG, Republic of Tajikistan Coal Sector Development Concept until 2040

With all above in mind, the question arises **whether under such conditions is it possible to achieve the increase in coal extraction targets that have been set in Tajikistan?** In accordance to the forecast developed for the purpose of this study an unambiguous answer is **no, it is not**. It is because according to the calculations presented in this paper, in 2030 the cumulative demand for Tajikistan's coal by the domestic and foreign buyers will be nearly 5 times less than anticipated in the *Coal Sector Development Concept*. In 2040, in turn, it will be 6 times less (see **Figure 9.1**). A gap that makes the said targets unachievable.

The analysis of the current status and development prospects of the coal market in Tajikistan discussed in this study leads to the following conclusions and recommendation:

- The power sector still has a cost-effective option of constructing a new coal-fired power plant. However, this option can be funded only from the national sources, which in practice cannot be mobilized, primarily due to a high debt burden of OSHC "Barki Tojik".
- The power sector development in Tajikistan should be based on the construction of power plants using renewable energy sources (sun and wind) and HPP, as such projects can be developed and executed in cooperation with, and with the support of the international development partners.
- While this option will eliminate the possibility to benefit from the abundant coal reserves in Tajikistan, the clean energy sources, if substituted for coal in power generation, will in the long-term perspective improve the competitiveness of the local products (particularly if the CBAM application becomes widespread), facilitate achievement of the climate neutrality, and allow to avoid significant costs related to public health.
- In the manufacturing industry, provision of any support for coal utilization in lieu of gas would be counterproductive, as it would add to the cost of final products and thus undermine their competitiveness. At the same time, where the use of coal is currently dictated by the economic calculation (as it is the case e.g., in the cement industry) it should not be immediately and arbitrarily prevented by any prescriptive measures, but the incentives for searching for alternative solutions should be created. In the long-term, any companies which products are currently based on power or heat generated by coal are likely to become uncompetitive if they do not modernize their mode of production, so that it aligns with the principles of the green economy.
- The projects using coal as feedstock (e.g., in the iron and steel industry, or in coal-to-olefins conversion processes) should not be completely ignored. However, such projects shall in no case follow suit of the failed Tajik Iron and Steel Works construction project. While in many cases an access to cheap coal might be an important business success factor, it is always only one of many variables, and therefore any potential coal-related project shall rely upon a thorough feasibility study, which shall be as accurate and unbiased as possible.
- The situation that occurred in the global markets in 2022 was extremely favourable for Tajikistan's coal export, but by no means should it be assumed to become a "new normal". As long as the price situation permits, the coal export shall naturally be continued, but the decisions to invest into any additional export infrastructure (such as e.g., construction of railroads particularly for that purpose) or expansion of the extraction capacities shall be made within the context of the real prospects for the global coal market development, which strongly indicate the future continuous decrease in both prices and demand. The Tajikistan's goals to increase coal extraction stipulated in the Governmental strategic documents should be deemed outdated. Due to the rapidly changing global situation, the goals set in 2016 are currently unattainable.
- In addition, *the Republic of Tajikistan Electric Transport Development Program for 2023-2028* and *the Republic of Tajikistan "Green" Economy Development Strategy for 2023-2037* adopted

at the end of 2022 contradict the coal sector development strategy. While, according to the *Coal Sector Development Concept*, the dependence on the import of petroleum products should be reduced by building coal-to-petroleum plants, *the Republic of Tajikistan Electric Transport Development Program for 2023-2028*, as indicated by its title, envisages an alternative solution to the problem.

- A new coal sector development strategy shall not be focused on achieving the record-high extraction levels, but rather on fair distribution of the natural resource rent obtained from coal export, ensuring safe working environment, mitigation of environmental damage (e.g., through decommissioning of the exhausted mines), and monitoring of methane emissions along the whole coal value chain. Provisions for the transformation and adaptation of local communities, jobs and economic activities dependent on coal extraction should be considered at the outset so as to mitigate the economic and social impacts of mine closure.

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Annexes

Annex A. Coal prices in Tajikistan

Table A.1 - Information on the cost of selling 1 tonne of coal, including VAT, in coal sector enterprises in the period 2022, TJS

№b /t	Enterprises	The cost of 1 tonne of coal				Average cost per tonne of coal
		Low quality coal	Mixture	Powder mixture	Lump coal	
1	Subsidiary "Fon-Yaghnob Mine"	-	304.99	327.50	350.00	327.50
2	Talco Resources Limited Liability Company	75.00	257.33	273.33	314.33	236.25
3	Closed Joint-Stock Company Combinati Metallurgii Tojik	-	300.00	210.00	330.00	280.00
4	Open Joint Stock Company Angisht	-	300.00	502.00	1 000.00	600.66
5	Subsidiary Ziddi Coal Field	-	130.00	150.00	210.00	163.33
6	Vena Limited Liability Company	310.00	375.00	395.00	455.00	383.75
7	Dion Angisht Tojikiston Open Joint Stock Company	-	-	292.97	-	202.97
8	Open Joint Stock Company "Kudrat 2010" - open	-	-	-	138.00	138.00
9	Open Joint Stock Company "Gunch & Co"	-	-	150.89	-	150.89
10	Open Joint Stock Company Angishti Kamarob	-	354.55	444.93	614.13	471.20
11	Open Joint Stock Company Siyovush-F - underground	-	-	300.18	-	300.18
12	Daryo 2015 Limited Liability Company	-	-	259.00	-	259.00
13	Anavac Limited Liability Company	-	-	163.46	-	163.46
14	Shirkati Tichoratiya Sayyod Limited Liability Company	-	100.58	108.25	-	104.42
	Average price of coal in Tajikistan	192.50	231.00	275.19	426.43	270.12

Source: SUE Angishti Tojik

Table A.2 - Information of the cost of selling of 1 tonne of coal, excluding VAT, at coal enterprises in the period 2022, TJS

№b /t	Enterprises	The cost of 1 tonne of coal				Average cost per tonne of coal
		Low quality coal	Mixture	Powder mixture	Lump coal	
1	Subsidiary "Fon-Yaghnob Mine"	-	265.21	284.78	304.35	294.78
2	Talco Resources Limited Liability Company	65.22	223.76	237.68	273.33	200.00
3	Closed Joint-Stock Company Combinati Metallurgii Tojik	-	260.87	182.61	286.96	243.48
4	Open Joint Stock Company Angisht	-	260.87	436.52	869.56	522.32
5	Subsidiary Ziddi Coal Field	-	110.17	127.12	177.97	138.42
6	Vena Limited Liability Company	269.56	326.09	343.47	395.65	333.69
7	Dion Angisht Tojikiston Open Joint Stock Company	-	-	254.76	-	254.76
8	Open Joint Stock Company "Kudrat 2010" - open	-	-	-	120.00	120.00
9	Open Joint Stock Company "Gunch & Co"	-	-	127.87	-	127.87
10	Open Joint Stock Company Angishti Kamarob	-	308.30	386.89	534.03	409.74
11	Open Joint Stock Company Siyovush-F - underground	-	-	261.03	-	261.03
12	Daryo 2015 Limited Liability Company	-	-	225.22	-	225.22
13	Anavac Limited Liability Company	-	-	142.14	-	142.14
14	Shirkati Tichoratiya Sayyod Limited Liability Company	-	87.46	94.13	-	90.79
	Average price of coal in Tajikistan	167.39	230.34	238.79	370.23	240.30

Source: SUE Angishti Tojik

Table A.3 - Average price of coal by grade, excluding VAT, TJS/t

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	8 months 2022
Price of coal per tonne	132.23	167.20	197.64	213.37	195.89	187.06	191.79	171.90	188.07	216.04	235.73
Bituminous coal	132.46	171.13	188.47	203.28	216.04	196.66	203.84	203.84	177.86	203.77	224.07
Lignite	130.83	168.50	229.73	248.68	170.82	153.47	155.63	155.63	243.25	289.67	294.08
Anthracite											

Source: Agency on Statistics under the President of the Republic of Tajikistan

Annex B. Location and characteristics of coal deposits in Tajikistan

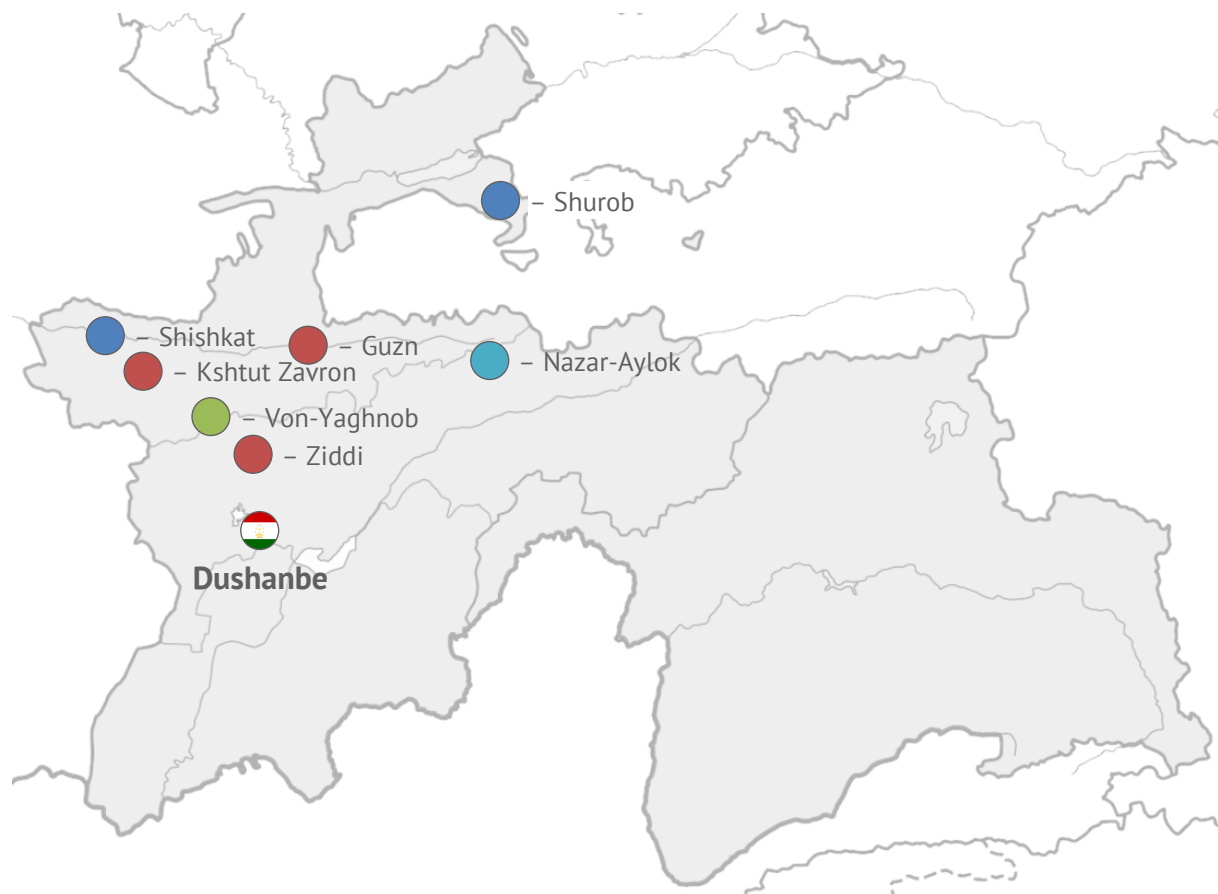
Table B.1 - Characteristics of the main coal deposits in Tajikistan

Field	Type of coal	Coal production in 2021, thousand tonnes	Commercial reserves, mln tonnes (categories A+B+C1, 01.01.2019)	Share in production, %	Share in stocks, %
Fon-Yaghnob	Bituminous coal, including coking coal	1 569	184	75%	62%
Ziddi	Bituminous coal	281	20	13%	7%
Shurob	Lignite	91	79	4%	26%
Nazar-Aylok	Anthracite	61	11	3%	4%
Guzn	Bituminous coal	60	0	3%	0%
Kshtut Zavron	Bituminous coal	10	0	0%	0%
Shishkat	Lignite	5	1	0%	0%
Others	Bituminous coal	12	4	1%	1%
Total		2 089	299	100%	100%

Source: data provided by a national consultant

Figure B.1 - Location of major coal deposits in Tajikistan

● – Lignite ● – Bituminous coal ● – Bituminous coal, including coking ● – Anthracite



Source: data provided by a national consultant

Annex C. Reserves of coal deposits in Tajikistan

Table C.1 - Resources of coal industry enterprises in Tajikistan

№	Field	Section	Enterprise	№ of protocol and year of approval	Approved reserves by the State Reserves Commission (GKZ), thousand tonnes			Company's resource balance as of 01.01.2019, thousand tonnes			
					A+B+C1	C2	A	B	C1	A+B+C1	C2
1	Shurob	№8 and 1/2	OJSC Angisht	№1 GKZ USSR 1992	103 581.3	55 402.0	7 658.6	23 676.0	47 570.0	78 904.6	55 402.0
2	Fon-Yaghnob				189 824.0	164 714.9		35 718.5	148 684.4	184 392.8	159 656.0
		Careerny	SE Fon Yaghnob Mine	№10946 GKZ USSR 1990	29 906.0	3 726.0	-	16 562.2	10 125.0	26 677.2	3 286.0
		Kante	Talco Resources Ltd.	№10360 GKZ USSR 1988	25 602.0	5 318.9	-	12 939.0	11 634.8	24 573.8	5 262.0
		Digicrud		№5 CDC of the USSR Ministry of Geology and Mineral Resources 1988	61 490.0	75 176.0	-	1 114.0	59 949.7	61 063.7	75 176.0
		Rawot	C.M.T. Ltd.	№5 CDC of the USSR Ministry of Geology and Mineral Resources 1988	72 826.0	75 932.0	-	5 103.3	66 974.9	72 078.1	75 932.0
3	Ziddi				21 173.0	25 531.0	-	7 595.3	12 582.5	20 177.8	25 531.0
		Zapadny	SE UM Ziddy	№1 the Bainidorav Commission 1970	20 953.0	25 069.0	-	7 595.3	12 511.0	20 106.3	25 069.0
		Vostochny		№90 KDZ RT, 2013	220.0	462.0	-	-	71.5	71.5	462.0
4	Nazar-Aylok				3 669.4	36 272.6	-	386.9	10 606.3	10 993.2	284 517.2

№	Field	Section	Enterprise	№ of protocol and year of approval	Approved reserves by the State Reserves Commission (GKZ), thousand tonnes			Company's resource balance as of 01.01.2019, thousand tonnes			
					A+B+C1	C2	A	B	C1	A+B+C1	C2
		Shikor-Hona	UM Nazar-Aylok SE	№32 KDZ RT, 1995	2 864.0	1 774.0	-	386.9	9 882.0	10 268.9	54 255.0
		Kaftar-Hona	Kamarob Ltd.	NTS Saridorai Geology	805.4	230 262.0	-	-	724.3	724.3	230 262.2
5	Sayyod		S.T. Sayyod Ltd.	№ 84 KDZ RT, 2013	427.9	47.0	-	-	246.1	246.1	47.0
6	Miyonadu		Wuromoon Ltd.	NTS UTGR, 1995	2 252.0	2 531.3	-	-	2 252.0	2 252.0	2 547.4
7	Mogiyon	Gesan	Dion Angisht Ltd.	№ 112 TKZ Taj UG, 1956	1 516.0	37 443.0	-	128.2	1 349.0	1 477.2	37 431.2
8	Guzn	Guzn	Kudrat 2010 Ltd.	№37VKZ M,P RT, 2008	254.2	13 141.0	-	133.4	112.3	245.7	13 117.4
9	Shishkat		Siyovush F Ltd.	NTS Tajikgeology 1990	793.8	2 138.5	-	170.4	621.3	791.7	2 138.5
9	Chashmasang		Gunch & Co Ltd.	№31 CDC TGH 1995	-	210.0	-	-	-	-	29.4
10	Shuroobod		SE UM Shuroobod	NTS SUTGRE, 1989	-	88.3	-	-	-	-	62.2
11	Toshkutan		DP UM Istiqlol	№1 the Bainidorav Commission 1970	Forecast	-	-	-	-	-	-
12	Hakimi		Anavac Ltd.	№10 CDC Tajgeo, 1993	-	116.0	-	-	-	-	102.6
13	Ravnou		VTI Pomir Ltd.	№1 the Bainidorav Commission 1970	Forecast	-	-	-	-	-	-
Total					322 697.8	330 934.2	7 658.6	67 808.7	224 023.9	299 481.1	580 581.9

Source: data provided by a national consultant

Annex D. Coal mining in Tajikistan

Table D.1 - Coal production in Tajikistan for 2007-2015, thousand tonnes

Name of enterprises	Company affiliation	2007	2008	2009	2010	2011	2012	2013	2014	2015	2007-2015
Sughd province		53.7	68.7	70.7	81.3	120.0	211.2	354.1	674.2	804.5	2 438.4
OAo Angisht (Shurob coal field)	State-owned	22.5	28.1	30.0	30.2	32.0	42.8	51.5	56.2	57.5	350.8
Fon-Yaghnob Mine (Fon-Yaghnob coal field)	State-owned	25.0	33.4	34.1	44.1	80.1	160.0	295.0	477.1	600.1	1 749.0
OOO TalkoResource (Fon-Yaghnob coal field)	Private	0.0	0.0	0.0	0.0	0.0	0.0	0.0	132.7	139.1	271.8
Anzob JV, Ugolny site (Fon-Yaghnob coal field)	Private	4.3	4.8	4.4	4.2	4.9	4.6	1.6	2.0	1.7	32.5
DP Maghiyan (Dion Angishti Tojikiston Ltd)	Private	1.9	2.0	1.5	2.2	2.3	3.0	3.2	3.3	3.4	22.8
OOO Guzn (Guzn coal Field)	Private	0.0	0.5	0.6	0.7	0.7	0.7	0.5	0.6	0.6	4.9
OOO Siyovush-F (Shishkat coal field)	Private	0.0	0.0	0.0	0.0	0.0	0.1	2.3	2.3	2.0	6.6
Districts under Central Government Jurisdiction		108.1	126.6	101.9	119.0	111.1	197.9	162.5	198.9	234.2	1 360.2
DP "Ziddi" (Western part of coal field Ziddi)	State-owned	45.1	35.5	13.0	27.8	23.8	46.2	45.4	81.7	116.9	435.4
SE "Nazar-Ailok" (Nazar-Aylok coal field)	Private	16.9	18.0	4.0	8.1	17.3	17.4	30.5	31.0	52.0	195.3
DP Toshkutan (Sangitang coal field)	Private	1.3	0.7	0.0	1.0	1.0	1.2	0.7	1.2	1.2	8.3
SHT Sayyod LLC (Sayyod coal field)	Private	27.2	34.4	30.8	27.5	25.0	28.0	26.2	30.0	34.0	262.9
LLC ANAVAK (Hakimi coal field)	Private	1.5	0.3	0.5	0.0	0.0	0.0	5.5	2.1	1.3	11.2
Ganch Ltd (Chashma-Sang Coal Field)	Private	2.5	4.0	1.4	0.0	0.0	0.0	0.0	1.0	1.1	9.9
Woromun Ltd. (Miyonadu coal field)	Private	5.1	5.4	2.1	2.2	0.7	1.5	1.3	2.0	2.0	22.2

Name of enterprises	Company affiliation	2007	2008	2009	2010	2011	2012	2013	2014	2015	2007-2015
Sanggalt Ltd (Eastern Part of coal field Ziddi)	Private	8.5	25.9	40.5	51.9	40.7	93.0	42.3	42.4	16.0	361.2
OOO Kamarob (Nazar-Aylok coal field)	State-owned	0.0	2.4	9.6	0.6	2.5	10.6	10.7	7.5	9.7	53.6
Khatlon region		2.7	30.1	3.1	3.1	3.1	1.7	0.8	1.0	2.7	48.3
DP Shuroobod (Shuroobod coal field)	State-owned	2.7	30.1	3.1	3.1	3.1	1.7	0.8	1.0	2.7	48.3
GBAO		0.2	0.8	0.8	0.0	2.5	1.1	0.4	0.6	1.1	7.6
LLC VTI-Pamir (Ravnov coal field)	Private	0.2	0.8	0.8	0.0	2.5	1.1	0.4	0.6	1.1	7.6
Total for Tajikistan		164.7	226.3	176.6	203.3	236.7	411.8	517.7	874.7	1 042.5	3 854.5
Plan for Tajikistan		120.0	125.2	170.0	184.0	200.0	220.0	232.0	710.0	700.0	2 541.2

Source: data provided by a national consultant

Table D.2 - Coal production in Tajikistan for 2016-2022, thousand tonnes

Name of enterprises	Company affiliation	2016	2017	2018	2019	2020	2021	2022	2016 - 2022
Sughd region		1 170.0	1 507.3	1 742.3	1 789.2	1 761.2	1 692.8	1 889.7	11 552.5
OAo Angisht (Shurob coal field)	State-owned	50.4	52.7	61.1	63.5	73.9	91.0	125.1	517.7
Fon Yaghnob Mine (Fon-Yaghnob coal field)	State-owned	641.7	789.0	882.9	889.2	862.3	736.1	650.1	5 451.3
OOO TalkoResource (Fon-Yaghnob coal field)	Private	354.5	402.0	435.8	412.2	435.8	455.6	573.2	3 069.1
Kombinati Metolurgii Tojik Ltd (coal mine "Fon-Yaghnob")	Private	117.7	257.0	346.6	400.0	376.9	343.2	426.7	2 268.2
Anzob JV, Ugolny site (Fon-Yaghnob coal field)	Private	1.5	0.0	0.0	0.0	0.0	0.0	0.0	1.5
DP Maghiyan (Dion Angishti Tojikiston Ltd)	Private	2.1	3.1	10.1	16.3	0.2	1.6	14.1	47.3
OOO Guzn (Guzn coal field) (Kudrat-2010)	Private	2.2	3.5	2.0	7.5	12.1	60.0	93.5	180.7
OOO Siyovush-F (Shishkat coal field)	Private	0.0	0.0	3.9	0.5	0.0	5.3	7.0	16.7
Districts under Central Government Jurisdiction		189.5	250.7	162.1	237.9	267.9	395.9	503.4	2 007.4
DP "Ziddi" (Western part of coal field Ziddi)	State-owned	66.8	88.6	103.8	195.6	200.5	280.5	300.7	1 236.6
SE "Nazar-Ailok" (Nazar-Aylok coal field)	Private	35.8	26.0	0.0	0.0	0.0	0.0	0.0	61.8
DP Toshkutan (Sangitang coal field)	Private	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.2
SHT Sayyod LLC (Sayyod coal field)	Private	37.6	43.4	4.8	1.3	3.4	2.8	23.5	116.8
LLC ANAVAK (Hakimi coal field)	Private	0.0	0.0	0.0	0.0	0.0	3.3	24.5	27.8
Vena Ltd (Fon Yagnob coal field)	Private	0.0	0.0	0.0	0.0	0.0	34.0	80.1	114.1
Daryo-2015 Ltd (Kshtut Zavron coal field)	Private	0.0	0.0	0.0	0.0	0.0	10.0	6.0	16.0
Ganch Ltd (Chashma-Sang Coal Field)	Private	1.4	1.7	4.0	4.0	4.0	4.0	7.5	26.6

Name of enterprises	Company affiliation	2016	2017	2018	2019	2020	2021	2022	2016 - 2022
Woromun Ltd. (Miyonadu coal field)	Private	2.1	2.1	1.2	0.0	0.0	0.0	0.0	5.4
Sanggalt Ltd (Eastern Part of coal field Ziddi)	Private	20.8	50.8	48.3	0.0	0.0	0.0	0.0	119.9
OAo Angishti Kamarob (Nazar-Aylok coal field)	State-owned	25.0	37.9	0.0	37.0	60.0	61.3	61.1	282.3
Khatlon region		2.1	2.2	2.6	0.7	0.0	0.0	0.0	7.5
DP Shuroobod (Shuroobod coal field)	State-owned	2.1	2.2	2.6	0.7	0.0	0.0	0.0	7.5
GBAO		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
LLC VTI-Pamir (Ravnov coal field)	Private	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total for Tajikistan		1 361.6	1 760.1	1 907.1	2 027.8	2 029.1	2 088.7	2 393.1	13 567.4
Plan for Tajikistan		1 700.0	1 800.0	1 900.0	1 950.0	2 000.0	2 100.0	2 160.0	13 610.0

Source: data provided by a national consultant

